Artificial Intelligence Review (2026) 59:13
https://doi.org/10.1007/s10462-025-11416-2

Machine learning powered financial credit scoring: a systematic literature review

Helmi Ayari[1] · Pr. Ramzi Guetari[1,2] · Pr. Naoufel Kraïem[3]

Received: 10 August 2024 / Accepted: 30 September 2025 / Published online: 18 November 2025
© The Author(s) 2025

Abstract
Over the past few decades, credit scoring has become an important tool in the financial sector. It enables banks and financial institutions to assess the creditworthiness of individuals and reduce the risk of default. As a result of significant advances in artificial intelligence techniques. Machine learning (ML) has made it possible to improve credit scoring by distinguishing between people with good creditworthiness and those with poorer creditworthiness. In this article, we propose a systematic literature review of ML-based financial credit scoring methods published between 2018 and 2024. A total of 330 research papers were extracted from four different online databases and digital libraries. After the study selection procedure, 63 research papers were selected for this systematic review. This paper aims to identify the major ML methods used in credit scoring, assess their strengths and limitations, and highlight notable trends and advancements. In addition, the review addresses the critical challenges faced in the adoption of ML models for credit scoring. This study not only contributes to the understanding of effective ML techniques used for credit scoring but also guides future research by highlighting the promising avenues in ML-based credit scoring efforts.

Keywords Credit scoring · Machine learning · Deep learning · Ensemble learning · Classification

✉ Helmi Ayari
helmi.ayari@ept.rnu.tn

Pr. Ramzi Guetari
r.guetari@novobit.ai

Pr. Naoufel Kraïem
nkraiem@kku.edu.sa

1 SERCOM Laboratory, Polytechnic School of Tunisia, University of Carthage, 2078 La Marsa, Tunisia

2 Novobit GmbH, Theodor-Heuss-Straße, 38122 Braunschweig, Germany

3 College of Computer Science, King Khalid University, 61421 Abha, Kingdom of Saudi Arabia

1 Introduction

In the contemporary highly dynamic and fast-paced financial landscape, banks and financial institutions face the challenge of processing a large volume of loan applications quickly and efficiently. Before the adoption of credit scoring, lending decisions were often influenced by subjective judgments, personal biases, and intuitive assessments. In addition, manual assessment of each borrower would be time-consuming and error-prone, leading to delays and inconsistent decision-making (Kumar et al. 2021).

Credit scoring has since emerged as a cornerstone of modern lending practices. By offering a systematic, data-driven approach to evaluating borrower creditworthiness, it has transformed the lending industry. At its core, credit scoring quantifies credit risk using applicants' financial behavior and repayment history. It ensures objectivity, minimizes human bias, and allows financial institutions to tailor loan terms based on risk profiles, thereby optimizing risk-adjusted returns (Atiya 2001). The automation of credit risk evaluation not only accelerates loan approval processes enhancing operational efficiency and customer satisfaction, but also strengthens financial stability for both lenders and borrowers.

Historically, standardized credit scoring models have been dominated by major credit bureaus such as Equifax (Kenny 2018), Experian (Bradford 2007), and TransUnion (Macey and Miller 1988). In the U.S., these agencies typically rely on the FICO score (Smith 2011), which assesses five key factors: payment history, amounts owed, length of credit history, new credit, and credit mix. These components are weighted to predict the borrower's likelihood of default.

Traditional models most commonly based on logistic regression or scorecards, are valued for their interpretability, transparency, and regulatory acceptance (Malmi 2001). However, their limitations are significant: they depend on narrow feature sets, assume linear relationships, and struggle to model complex behavioral patterns (Dastile et al. 2020). As a result, they often underperform with non-traditional or heterogeneous credit profiles.

Complementing these scoring frameworks are regulatory standards such as the International Financial Reporting Standard 9 (IFRS 9) (ElKelish 2021), which mandates the estimation of Expected Credit Loss (ECL) through three parameters: Probability of Default (PD), Exposure at Default (EAD), and Loss Given Default (LGD). PD, derived from historical repayment data, is the most influential factor (Bhatore et al. 2020), while LGD reflects the proportion of unrecovered assets and is calculated as one minus the recovery rate (Bhandary and Ghosh 2025). EAD denotes the loan exposure subject to credit risk. IFRS 9 promotes proactive credit risk recognition, enabling more accurate provisioning by integrating historical data, present conditions, and reasonable forecasts.

Recent advances in ML have unlocked new capabilities for credit risk modeling. ML models can capture nonlinearities, handle high-dimensional data, and adapt to evolving borrower behavior (Lenka et al. 2022). They also allow the incorporation of alternative data sources such as social media or mobile usage, enhancing prediction accuracy and enabling more personalized assessments (Markov et al. 2022). Consequently, ML has garnered increasing attention for its potential to improve the accuracy, fairness, and efficiency of credit scoring systems (Hayashi 2022).

Nevertheless, several research challenges persist. Many studies lack standardized datasets, preprocessing methods, or evaluation metrics, hindering cross-study comparisons. Complex models, such as Deep Learning (DL) and hybrid systems, often lack interpretabil-

ity, and although tools like LIME (Ribeiro et al. 2019) and SHAP (Lundberg and Lee 2017) are emerging, their reliability and adoption remain limited. Integrating alternative data sources into scalable systems raises ethical and privacy concerns. Issues of algorithmic bias remain insufficiently addressed, posing serious risks of discriminatory lending. Moreover, high-dimensional inputs can lead to overfitting or computational inefficiency, and feature selection is frequently overlooked. The deployment of advanced models is also hampered by their computational complexity, which may be prohibitive for smaller institutions.

This systematic literature review (SLR) presents a comprehensive synthesis of existing research on ML applications in credit scoring. It evaluates how ML methods address the limitations of traditional approaches and explores their capacity to deliver scalable, reliable, and ethically sound credit decisions. We identify key ML techniques, assess their strengths and weaknesses, and analyze emerging trends and innovations. This review also addresses critical challenges in implementation, including interpretability, bias, and dimensionality, and highlights areas for future research.

By consolidating findings from a broad range of studies, this SLR aims to inform researchers, practitioners, and financial institutions alike. It clarifies gaps in the current literature, summarizes major opportunities and risks, and offers actionable recommendations. Ultimately, this work contributes to the development of more transparent, fair, and robust ML-based credit scoring systems, advancing financial inclusion and reinforcing trust in automated financial decisions.

The rest of this paper is organized as follows: Sect. 2 lists some related works that overlap with the problem of credit scoring. Sect. 3 describes the methodology of the literature review. Section 4 presents the ML methods used for credit scoring. Section 5 presents the performance evaluation criteria. Section 6 presents the results. Section 7 presents the discussion of this review. Section 8 addresses the limitations of this review, and Sect. 9 concludes the paper with ideas for further work.

2 Related work

Several literature reviews have examined the use of ML techniques for borrower classification in credit scoring. This section summarizes the most important reviews in this area.

Dastile et al. (2020) examined 74 studies (2010–2018) on statistical and ML models for credit scoring. They found that ensemble methods such as Random Forests (RF) and Extreme Gradient Boosting (XGBoost) outperformed single classifiers and reaching 79% accuracy. Among DL models, Convolutional Neural Networks (CNNs) achieved the highest AUC of 90% on the German dataset and 99% on the Australian dataset. Key limitations identified included the lack of macroeconomic variables and insufficient exploratory data analysis, both critical for robust model development. The authors proposed a framework addressing data preparation, feature extraction, and baseline models like Logistic Regression (LR) and Decision Trees (DT). They recommended future work on class imbalance (like SMOTE Chawla et al. 2002), feature selection, and alternative classification thresholds.

In Kumar et al. (2021), the authors focused on credit scoring in rural finance, emphasizing the role of Financial Technology (fintech) and ML/AI in improving access to underserved populations. They highlighted ML models such as Artificial Neural Networks (ANN), SVMs, RF, LR, and hybrid models as effective tools for credit assessment. The study found

that hybrid and AI-ML-based models yielded higher accuracy and efficiency by integrating multiple techniques. Additionally, it stressed the importance of regulatory frameworks and algorithmic transparency to ensure ethical implementation. These models were especially impactful in rural contexts, enabling more inclusive credit access through the processing of diverse data sources.

Hayashi (2022) conducted a systematic review (2019–2022) on DL in credit scoring, highlighting the superior performance of Deep Belief Networks (DBNs) over models like SVMs (Noble 2006), Gradient Boosting Trees (Ke et al. 2017), RFs (Breiman 2001), and LR (Wright 1995). DBNs, using unsupervised learning, effectively extract deep features. The study compared DL with ensemble and hybrid models on standard datasets and explored novel approaches, such as converting tabular data into images for CNNs (Gu et al. 2018). It addressed the interpretability challenge through rule extraction and stressed compliance with GDPR (Tokarski 2020). The highest accuracy (98.66%) was achieved by Acharya et al. (2022), with another model scoring 93.16% on the Japanese dataset (Zhang et al. 2021). The review concluded that DL holds strong potential for integrating structured and unstructured data with improved accuracy and explainability.

Lenka et al. (2022) presented an analysis of Ensemble Learning (EL) for imbalanced credit scoring dataset. The research explored the impact of resampling techniques like SMOTE and different feature selection methods such as information gain, principal component analysis, and Genetic Algorithms (GAs) on improving the performance of credit scoring models. The study conducted an extensive comparative analysis of 5 bases and 14 ensemble models using german, australian, and japanese datasets. The results highlighted the effectiveness of the GA based feature selection technique and the CatBoost algorithm, achieving the best accuracy with 86.70%, 88.40%, and 86.20%, respectively. The study concluded by recommending the combination of CatBoost with GA-based feature selection for building accurate and reliable credit scoring models.

In Markov et al. (2022), a review of credit scoring methodologies from 2016 to 2021 was conducted, highlighting the evolution of credit risk assessment and its influence on lending, investment, and risk management decisions. Using a systematic approach, the authors compared recent trends with those from 1991 to 2015, examining the shift from traditional models such as LR and DTs to more advanced techniques like SVMs, ensemble methods, and neural networks. A key finding was the growing prominence of ensemble models, recognized for their superior predictive performance. The review also noted the diversity of complex models grouped under the "Other" category, which, despite performance variability, often produced strong results when appropriately applied. Furthermore, the study highlighted the frequent use of public datasets such as the Australian and German credit datasets, which enhance model development, address class imbalance issues, and support reproducibility and generalization. This reflects a broader trend in credit scoring research.

Kamimura et al. (2023) conducted a review of 46 studies on optimization methods for Credit Scoring Models (CSMs) published between 2008 and 2022. The review identified a wide use of techniques including financial analysis, ML, and data mining. Logistic Regression (13%), Naive Bayes (10%), and Neural Networks (7%) were the most commonly used methods, with a growing trend toward hybrid models (72%). The study emphasized the need to integrate big data and DL for future CSM development and highlighted the importance of addressing legal, ethical, and practical issues. It also called for more research focused on small businesses and the use of diverse data sources.

3 Methodology

3.1 Protocol registration and guidelines

This SLR adheres rigorously to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al. 2021; Bhandary et al. 2024). Although no formal protocol registration (like PROSPERO) was undertaken, the methodology was defined in advance to ensure transparency, reproducibility, and reduce potential bias.

3.2 Ethical considerations

This study is a SLR and did not involve any direct interaction with human participants or the collection of primary data. Therefore, ethical approval and informed consent were not applicable. All data included in this review were obtained from previously published studies, which are assumed to have followed the appropriate ethical standards and obtained informed consent from their participants. In conducting this review, we adhered to the PRISMA 2020 guidelines to ensure methodological transparency, reproducibility, and integrity.

3.2.1 Research questions

The research questions serve as a foundation for the SLR, guiding the design, data collection, and synthesis phases to ensure the review remains focused and relevant. The research questions formulated for this study are presented below:

RQ1: What are the most widely used ML models for credit scoring?
RQ2: What are the strengths and limitations of ML models used for credit scoring?
RQ3: What metrics are used to evaluate ML credit scoring models?
RQ4: What are the emerging trends and advances in ML models for credit scoring?
RQ5: What are the challenges in adopting ML models for credit scoring?

3.3 Information sources and search strategy

3.3.1 Database selection

Relevant literature was retrieved from four major digital libraries: 'Springer Link', 'ACM digital library', 'IEEE Xplore', and 'Google Scholar' (Table 1). The search was performed using the full text of the papers available in these databases.

Table 1 Selected digital libraries

Digital libraries	URL
SpringerLink	http://link.springer.com
ACM Digital Library	http://dl.acm.org
IEEE Xplore	http://ieeexplore.ieee.org
Google Scholar	http://scholar.google.com

3.3.2 Search strategy development

The search strategy is derived by selecting specific keywords and their synonyms from the identified research questions. These keywords are then organized in a specific order using the 'AND' and 'OR' operators to construct the following query:

```
("credit scoring" OR "credit assessment" OR "credit prediction") AND ("Data Mining" OR "Artificial Intelligence" OR "AI") AND ("machine learning" OR "ML") AND ("classification" OR "classifier" OR "predictive modeling" OR "algorithm" OR "method" OR "technique" OR "model") AND ("deep learning" OR "DL") AND ("hybrid models" OR "novel models") AND ("supervised learning" OR "unsupervised learning" OR "ensemble learning")
```

3.3.3 Temporal scope

The search encompassed publications from January 2018 to December 2024, capturing the period of significant ML advancement in financial services while ensuring contemporary relevance.

3.4 Eligibility criteria

This review applied specific eligibility criteria to ensure the relevance and quality of the selected studies. The inclusion and exclusion criteria were defined prior to the screening process and were consistently applied throughout.

3.4.1 Inclusion criteria

Studies were included if they met the following conditions:

- Published between 2018 and 2024, to capture the most recent advancements in ML and their application to credit scoring.
- Written in English, to ensure interpretability and consistency in analysis.
- Focused on the use of ML techniques in credit scoring, specifically studies that investigated their application to predicting creditworthiness or default risk.
- Presented empirical results, particularly those involving experiments on publicly available or clearly defined datasets.
- Published in peer-reviewed journals or conference proceedings, to ensure academic quality.
- Addressed at least one of the predefined research questions outlined in Sect. 3.

3.4.2 Exclusion criteria

The following types of studies were excluded from the review:

- Studies published before 2018, as they fall outside the scope of recent developments in ML for credit scoring.

- Duplicate publications or articles with overlapping content.
- Papers with fewer than four pages, which often lacked sufficient methodological depth.
- Studies that were incomplete, missing results, or lacked methodological transparency.
- Articles that did not provide clearly defined evaluation metrics for assessing model performance.
- Studies that did not address any of the research questions defined for this systematic review.

3.4.3 Study selection process

Following the PRISMA 2020 guidelines (Frank et al. 2024), the study selection process was conducted in four sequential phases: identification, screening, eligibility, and inclusion discussed in Sect. 6.1.

3.5 Data extraction process

3.5.1 Data extraction framework

To ensure consistency and completeness during the review process, a structured data extraction form was developed and implemented as an Excel spreadsheet. This form was designed to systematically collect relevant information from each selected study based on the predefined research questions. The extracted data focused on the following elements:

- ML techniques: the specific models applied to credit scoring, including traditional ML algorithms, DL models, and ensemble methods.
- Datasets: the names and sources of the datasets used in the experiments, as well as any descriptions provided regarding their nature or origin.
- Evaluation metrics: the performance measures used in the studies, such as accuracy, precision, recall, F1-score, AUC, and specificity.

This data extraction process allowed for a structured comparison between studies and served as the foundation for the synthesis phase. Each extracted element contributed to answering the research questions and identifying patterns, strengths, limitations, and common practices in ML-based credit scoring.

3.5.2 Quality assessment

Quality assessment criteria are used to determine the suitability of research papers to effectively address the research questions. Each question is assessed using the options “yes”, “partly”, and “no”, corresponding to values of 1, 0.5, and 0, respectively. Each paper could receive a maximum score of 5, with the total reflecting the overall quality. The list of quality assessment questions is provided below:

1. Does the article clearly and explicitly state the aims of the study?
2. Does the selected study provide sufficient information and details for performance assessment?

3. Do the references used in the research appear to be appropriate and to be adequate in terms of support for the study?
4. Does the study clearly and explicitly report the results obtained and the conclusions drawn?
5. Does the article provide background information that is relevant and appropriate to the topic?

The final score is obtained by summing the scores for all the quality assessment questions. Following the quality assessment, each selected study achieved a score of at least 77%. This score balances the inclusivity and selectivity of the SLR and focuses on robust methodologies that provide insights into ML for credit scoring. The 77% threshold serves as a pragmatic benchmark, ensuring both methodological rigor and adequate coverage.

4 Machine learning methods for the credit scoring process

ML, a subfield of AI, focuses on designing algorithms that enable systems to learn from data and make predictions or decisions without being explicitly programmed. These algorithms identify patterns in data and adapt their performance over time. ML can be broadly categorized into supervised learning (Caruana and Niculescu-Mizil 2006) and unsupervised learning (Dike et al. 2018).

In supervised learning, models are trained using labeled data, where each input (X) is associated with a corresponding output (Y). The model learns to map inputs to outputs, enabling it to classify or predict outcomes for new, unseen data. In contrast, unsupervised learning uses unlabeled data, where the model identifies patterns, similarities, or clusters without predefined output labels.

Various ML algorithms, such as conventional ML, DL, EL, and hybrid models, have gained significant attention due to their enhanced predictive capabilities in credit scoring tasks. The following sections explore the application and performance of these techniques in the context of credit scoring.

4.1 Traditional machine learning models

This section describes the conventional ML models used for credit scoring.

4.1.1 Logistic regression

LR is a well-established algorithm commonly used in credit scoring (Dastile et al. 2020). It is a probabilistic classification model that estimates the probability of a binary outcome (Atiya 2001). It is particularly valuable in credit scoring as it estimates the conditional probability of an input belonging to a particular class (default or non-default) (Ala'raj et al. 2022). Given a feature vector $\mathrm{x} \in \mathbb{R}^d$, the model computes (formula 1):

$$P(Y = 1|\mathrm{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathrm{x})}} \tag{1}$$

where $Y \in \{0, 1\}$ is the binary response variable (1 = default, 0 = non-default), β_0 is the intercept term, and $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_d)^T$ is the coefficient vector. The parameters $\beta_0, \boldsymbol{\beta}$ are estimated by maximizing the log-likelihood function:

$$\ell(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{N} [y_i \log p(\mathrm{x}_i) + (1 - y_i) \log(1 - p(\mathrm{x}_i))] \tag{2}$$

where $p(\mathrm{x}_i) = P(Y = 1|\mathrm{x}_i)$. This optimization is typically performed using iterative methods like Newton–Raphson or gradient descent (Hosmer and Lemeshow 2000).

Its simplicity, good performance with smaller datasets, and robustness against noise make it a favored choice for credit scoring. However, the assumption of a linear relationship between inputs and log odds may not always hold, especially in cases where the relationship is nonlinear.

Cao et al. (2021) introduced a ML credit score and default probability model with LR showing the best performance in 5-fold validation. They utilized attribute weighting for feature selection based on information ranking and recommended focusing on recall for imbalanced data. By setting an optimal probability threshold of 0.18 using Youden's index, they improved TPR while reducing FNR achieving 86.58% accuracy.

Dumitrescu et al. (2022) introduced Penalized Logistic Tree Regression (PLTR), a powerful and interpretable credit scoring system. They enhanced LR using rules from short-depth DTs as predictors and enabled non-linear effects capture while retaining interpretability. The proposed method achieved the best AUC with 92.99%, 77.80%, and 90.11% in autralian, taiwan, and housing datasets respectively. The proposed PLTR improved credit risk prediction of monte carlo simulations and real-world applications.

Ariza-Garzón et al. (2020) compared the LR model and other algorithms such as RF, DT, and XGBoost for peer-to-peer lending scoring. XGBoost excelled in the default class though it ranked third in the non-default class with the highest precision. The study revealed that LR achieved the best accuracy (78.10%) and AUC (66.60%) on the lending club dataset compared to the other ML models. Their research demonstrated the possibility of building accurate and transparent ML models. These models can win the trust of industry players, regulators, and end-users, especially in contexts where explainability is crucial in P2P lending.

4.1.2 Decision trees

DTs (Safavian and Landgrebe 1991) are supervised models used for both regression and classification tasks. The structure of the model mirrors that of a tree, with a root, branches, and leaves representing decisions made based on attribute values. DTs split data by selecting attributes that maximize information gain, forming a decision path until no further splits are possible (Aniceto et al. 2020). A key advantage of DTs is their interpretability, as they provide clear, understandable decision rules. They are also capable of capturing nonlinear relationships between features and the target variable, making them suitable for complex datasets. However, DTs can be highly sensitive to the training data, which leads to overfitting or instability when the dataset changes slightly (Ariza-Garzón et al. 2020).

Syed Nor et al. (2019) developed a personal bankruptcy prediction model using the DT technique. The study defined bankruptcy as terminated members who failed to settle their loans, using a sample of 24,546 cases with 17% settled and 83% terminated. The dataset included a dependent variable (bankruptcy status) and 12 predictors collected from an authorized debt management agency. The findings provided profiles of bankrupts, a reliable personal bankruptcy scoring model achieving 83.29% accuracy, 06.62% specificity, and 99.00% sensitivity, and identified significant variables on imbalanced data.

Khedr et al. (2021) developed a new predictive method for default customers' loans using ML. The method utilized available personal data and historical credit data to evaluate the creditworthiness of customers for loans. The ABE dataset was used for training and testing, incorporating 10 features from the application form and i-score report class to assist credit officers in making informed decisions and avoiding random customer selection. The performance of several classifiers was compared before and after feature selection. The results indicated that the DT classifier outperformed other classifiers with a significant prediction accuracy of almost 94.85% accuracy and 96.75% F1-score.

Maharjan (2022) applied three different classifiers, C4.5, CART, and Naïve Bayes, to predict loan grants and attribute selection. Their research aimed to help financial institutions seek better strategies through credit scoring models. They concluded that categories 4 and 8 were the best, while categories 3, 6, and 11 were the worst, as these had higher false positive values in all C4.5, CART, and Naïve Bayes testing. Among the classifiers, C4.5 was the best for predicting loans, achieving an 85.23% F1-score and 78.33% accuracy on the german loan dataset.

4.1.3 Random forest

RF is a supervised learning model that aggregates multiple DTs to make predictions for classification and regression tasks. The core technique behind RF is bagging, or bootstrap aggregation, which involves resampling the training dataset with replacement to create multiple subsets. A DT is trained on each of these subsets and predictions are made based on a majority vote across all trees (Dastile et al. 2020). RF is known for its speed and simplicity, effectively handling high-dimensional data, and being less prone to overfitting compared to individual DTs (Teles et al. 2021). However, the reliance of the model on numerous trees can complicate the interpretability of its results.

Trivedi (2020) developed a credit scoring prediction model through the integration of various feature selection methods such as information gain, gain ratio, and chi-square and ML classifiers such as naïve bayes, RF, DT (C5.0), and SVM. Their proposed model proved the effectiveness of a combination of RF and chi-Square in achieving strong performance with 93.12% accuracy and 93.10% F1-score on german dataset. However, it required a slightly longer training time than other approaches.

Li et al. (2021) introduced a deep forest or multi-grained Cascade Forest (gcForest) model based on a RF algorithm. The gcForest efficiently processes high-dimensional feature information using multidimensional scanning and cascading. They created a two-stage hybrid default discrimination model by combining gcForest with multiple feature selection methods. The proposed model achieved the best accuracy (81.20%) and AUC (86.80%) on a german dataset.

Moscato (2021) conducted a benchmarking analysis of prevalent credit risk scoring models for predicting loan repayment in P2P platforms. Addressing class imbalance, they evaluated classifiers with various sampling strategies and selected RF, LR, and Multi-Layer-Perceptron (MLP) combined with Random Under Sampling (RUS) and IHT. Among these, the RF-RUS combination demonstrated the most effective performance achieving 64.00% accuracy, 63.00% recall, and 71.70% AUC on the lending club dataset.

Aji and Dhini (2019) applied data mining to address non-performing loan issues focused on mortgage loans. Using a dataset from an Indonesian bank that provided mortgage loans, they used RF with AdaBoost classifiers. Their analysis indicated that the model achieved the best results with 72.95% accuracy, 73.00% recall, and 70.40% specificity.

Tran et al. (2021) developed a credit scoring method in vietnam. They employed machine-learning models such as LightGBM, CatBoost, and RF derived from their participation in the kalapa credit score challenge. Their evaluation ultimately determined RF as the best-performing model based on their experimental outcomes achieving 83.00% F1-score and 81.00% AUC on Kalapa dataset. In addition, this was the first model applied in the field of vietnamese banking.

Parvin and Saleena (2020) attempted to forecast credit scores using several classifier models (LR, RF, DT, SVM, K nearest neighbor, naive bayes, extra trees classifier, ada boosting, bagged DT, and MLP) and evaluated the effectiveness of each model using metrics. A comparison study was conducted to determine the best classifier for predicting credit scores. The experimental results showed that the RF model provided greater accuracy with 88.41%, recall with 80.00%, precision with 87.00%, and F1-score with 84.00% on the australian dataset.

Zhang et al. (2018) introduced a credit scoring model (NCSM) based on feature selection and grid search to optimize the RF algorithm. To improve prediction accuracy, the model lowered the influence of irrelevant and redundant information. Information entropy was regarded as the heuristic that was used to select the best feature in NCSM. The experimental results demonstrated that NCSM achieved 91.71% accuracy on the australian dataset and 82.14% accuracy on the german dataset.

4.1.4 Support vector machine

SVM is a supervised learning model that maps data into a high-dimensional space and separates it into two classes using a linear separator known as a hyperplane (Teles et al. 2021). Commonly used for classification tasks, SVM identifies the optimal hyperplane by maximizing the margin between classes while minimizing the distance of the closest points (known as support vectors) from the boundary (Cervantes et al. 2020). Various kernels, such as linear, polynomial, radial basis function, and sigmoid can be applied to handle different data structures (Friedman et al. 1997). With the appropriate kernel, SVM is effective for both linearly and non-linearly separable datasets. However, kernel selection significantly affects performance and may increase training time on large datasets (Huang et al. 2007).

Teles et al. (2021) applied credit scoring using collateral as an independent variable and compared SVM and RF models in forecasting the recovered credit value. The SVM model achieved a slightly higher classification accuracy (98.34%) compared to RF (98.20%) on a bank institution dataset.

Dm and Mm (2018) compared loan default prediction in Kenya using SVM and LR models. The data was used from equity bank and split into training and test sets. The LR model showed an accuracy of 77.27% with the train data and 73.33% with the test data, and a precision of 84.40% and 82.44%, respectively. The SVM with a linear kernel model had an accuracy of 88.29% and 86.12% with the train and test data, respectively, and a precision of 87.85% and 78.31%. The SVM model outperformed the LR model, which led the study to recommend the use of SVM for loan default prediction in financial institutions.

Wang and Li (2019) improved credit assessment predictions using an SVM-based model. Recognizing that SVM performance heavily relies on parameter selection, they employed an Improved Fruit Fly Optimization Algorithm (IFOA) to optimize these parameters. The study analyzed P2P loan data using Linear Regression, Classical SVM, FOA-SVM, and IFOA-SVM, finding that the IFOA-SVM model provided the most accurate predictions by achieving the best precision (93%).

4.1.5 K nearest neighbors

The K Nearest Neighbors (KNN) (Guo et al. 2003) model is a non-parametric supervised learning technique that functions by utilizing two parameters, the distance function and the selected k value, with performance based on the aforementioned factors. KNN initially calculates the distance between all data points and accumulates those that are near to it for any new data point. The algorithm uses a chosen distance function (such as Euclidean or Manhattan) to identify and group the nearest neighbors to the target data point. Next, it collects a specified number of data points that have the shortest distance between them all and classify them based on their distance. However, because KNN calculates a distance metric for every data point during classification, it incurs higher computational demands which can be perceived as a drawback in terms of computational cost (Zhang et al. 2017).

Mukid et al. (2018) reviewed the Weighted K-Nearest Neighbor (WKNN) method for credit assessment, considering the use of various kernel functions. The research utilized credit data from a private bank in indonesia. The results demonstrated that the gaussian kernel and rectangular kernel significantly improved the performance of the WKNN method. Specifically, the gaussian and rectangular kernels achieved an accuracy of 82.40% and a very high sensitivity of 99.34%, though the specificity was relatively low (11.11%), which indicated that the model strongly favored identifying positive cases.

Loo et al. (2023) predicted the risk of loan default using various ML algorithms (LR, DT, RF, KNN, SVM, and naïve bayes) and compared these algorithms to identify the most suitable one for predicting loan default risk. In addition, they assisted the decision-makers in approving or rejecting loan requests in india. Using a dataset from kaggle focused on loan applicants in India, they analyzed behavior to determine risk. KNN emerged as the best model, scoring the highest in all evaluation metrics (accuracy, recall, precision, F1-score) with a score of 89%.

Pratiwi et al. (2019) applied the pseudo nearest neighbor (PNN) method to identify prospective borrowers eligible for loan proposals. The study used historical credit data from a national bank in indonesia to focus on characteristics such as age, number of children, business duration, income, loan amount, and credit period. If a new borrower had characteristics similar to a good historical borrower, the loan proposal was approved; otherwise, it was refused. The k-NN method achieved the best classification with k = 1, resulting in

the smallest error of 1.89%. The best classification for PNN was with k = 13, yielding the smallest error of 20.75%. Overall, k-NN proved to be more accurate for credit classification than PNN.

4.1.6 Hybrid and composite machine learning models

Hybrid and composite ML models integrate multiple algorithms to enhance predictive performance in credit scoring. By combining supervised and unsupervised techniques, they overcome individual limitations and improve accuracy and robustness.

Unsupervised learning models are valuable in credit scoring for uncovering hidden patterns in data. When used in hybrid models for feature selection (Tripathi et al. 2018) and segmentation (Boughaci et al. 2021), they improve prediction accuracy and provide essential inputs for supervised models, enhancing overall performance and discriminative power. Various studies have been proposed to prove the effectiveness of hybrid ML models in the credit scoring domain.

Goh et al. (2020) proposed a hybrid model that integrated Harmony Search (HS) for both feature selection and hyperparameter tuning. They introduced a Modified HS (MHS), incorporating elitism and exploration-exploitation strategies to improve efficiency. The combination of MHS and RF (MHS-RF) achieved 87.38% accuracy on the australian dataset, offering improved explainability and computational efficiency (full results in Table 6).

Nalic et al. (2020) suggested a novel hybrid data mining model that combined feature selection and EL methods. They utilized various preprocessing techniques and five alternative feature selection algorithms integrating their results through innovative voting methods. The hybrid model using the IfAny voting method and the GLM + DT ensemble outperformed other classifiers by reaching 87.69% accuracy and 87.69% F1-score on a bosnia dataset.

Yao and Chen (2019) proposed a new hybrid RF-SVM ensemble model that used RF to select essential variables and ensemble methods to aggregate SVM as a robust classifier. The testing results indicated that the proposed model achieved the best accuracy on the australian dataset with 87.94%, 83.85% recall, and 92.10% AUC. This model demonstrated promising effectiveness and potential for application in credit scoring.

Zhang et al. (2021) devised a credit-scoring hybrid ensemble model that combined voting-based outlier detection and balanced sampling. Their approach aimed to enhance prediction accuracy by reducing noise impact during classifier training. They introduced a weighted voting mechanism for outlier detection and employed bagging-based balanced sampling to address class imbalance. The effectiveness of the model was proved through experiments by achieving 99.77% and 99.71% F1-score on the creator dataset.

Tripathi et al. (2019) developed a hybrid model that improved credit scoring prediction through feature selection and a multilayer ensemble classifier framework. Their approach involved three phases: ranking and weighting classifiers, ensemble feature selection, and using selected features in a multilayer ensemble classifier architecture. Additionally, they introduced a Choquet integral-based classifier placement algorithm, achieving 92.69% accuracy, 97.16% recall, and 88.46% specificity on the australian dataset.

Using unsupervised learning, Yuan et al. (2022) introduced a two-stage default prediction model that combined k-means clustering for sample partitioning and support vector domain description (SVDD) for credit scoring. The model utilized data from multi-temporal

data and demonstrated a five-year default prediction capability (AUC >0.85). The results (Table 6) showed that the proposed model achieved the best results on the real-world dataset achieving 86.33% AUC and 86.12% G-mean.

Boughaci et al. (2021) developed a hybrid method using clustering and RF techniques for credit scoring and financial bankruptcy prediction. They employed k-means clustering to group applicants and then applied RF to the clustered data. The results showed that the proposed model achieved the best results on taiwan dataset with 100% recall, 100% precision, and 100% F1-score. The approach improved classification performance and showed promise in applicant segmentation.

Suleiman et al. (2021) proposed a method for improving the discriminant capabilities of KNN and neural networks using unsupervised learning based on a SOM. The knowledge obtained by SOM was used as input to the subsequent pattern recognition models in the two-stage method. The results showed that the two-stage models improved the performance of both neural networks and KNN performances achieving 96.30% accuracy with SOM with KNN while SOM with neural networks achieved 97.30% accuracy on the bank of agriculture dataset.

Bao et al. (2019) proposed a unique strategy that combines unsupervised and supervised learning for credit risk assessment. They applied unsupervised techniques in consensus models like SOM at two stages: consensus formation and dataset clustering using k-means to group samples by presence conditions. Their approach outperformed other methods achieving 92.00% accuracy on the chinese P2P credit dataset.

Ibrahim and Olagunju (2022) introduced a two-stage credit scoring model employing SOM and CART. This approach fed the knowledge from clusters of SOM into CART for classification. Results from BOA's Sokoto data showed enhancement and boosted the performance of CART from 96.30% to 96.70%. This integration of SOM with CART outperformed the standalone CART model.

4.2 Deep learning models

DL (Hayashi 2022) is a subset of ML that is based on ANN with multiple layers between the input and output layers. These deep neural networks are powerful tools since they are capable of extracting intricate patterns and features from complex datasets automatically. This section describes DL models used for credit scoring.

4.2.1 Artificial neural network

ANN are computational DL models inspired by biological neurons. They consist of interconnected artificial neurons that can perform complex computations and adapt their structures based on external signals or information. A common architecture is a MLP consisting of an input layer, one or more hidden layers, and an output layer. The training focuses on minimizing the loss function and refining predictions through weight and bias adjustments via backpropagation (Rumelhart et al. 1986). Despite their ability to automatically extract features and perform well on larger datasets, ANNs are considered a "Black Box" posing challenges in interpretation which is a crucial factor in credit scoring where transparency is vital (Ariza-Garzón et al. 2020).

Kazemi et al. (2023) proposed a method using a GA and neural networks to find optimal cut-off values based on performance metrics and the dataset. Their approach outperformed the standard threshold of 0.5 by achieving the highest accuracy with 91.91% and AUC with 92.60% on the australian dataset which resulted in more accurate classifications.

Kazemi et al. (2021) introduced a hybrid method using a GA to optimize the structural parameters of a neural network classifier for enhanced accuracy. They applied this approach to the australian and german credit scoring datasets and achieved significant improvements: 2.68% and 0.1% enhancements, respectively. The statistical analysis supported the effectiveness of their algorithm in parameter tuning.

Diaconescu and Neagoe (2020) introduced a credit-scoring DL technique that defined credit score as the weighted sum of false negative errors and false positive mistakes. This model aimed for the lowest possible score emphasizing FN (low credit projected as good) to minimize the missed alarm rate across the total number of faults. Optimization methods were employed to select a deep-learning neural network architecture and hyperparameters for their approach and the model achieved 84.83% accuracy on the german dataset.

4.2.2 Convolutional neural networks

CNNs are deep feedforward neural networks that have been widely used for their great capacity to extract local image information. Unlike traditional supervised learning methods, CNNs eliminate the need for manual feature extraction. They handle feature extraction and description during the learning phase and aim to minimize classification errors. The architecture comprises convolution, pooling, and fully connected layers, each with specific processes for visual property extraction and classification (Rumelhart et al. 1986). The convolution layer uses filters for feature detection, pooling reduces image size while preserving key characteristics and the fully connected layer performs classification on flattened features. While CNNs are standard for images, their direct application to structured credit data requires careful adaptation.

Dastile and Celik (2021) proposed an interpretable DL model for credit scoring to meet legal decision-making criteria. They transformed tabular data into images employing 2D CNNs. In their approach, each image pixel represented a feature from the tabular dataset. Their model was tested on three public credit scoring datasets achieving the best results on the australian dataset with 95.00% accuracy.

Zhu et al. (2018) proposed a model that combines CNN and the feature selection algorithm relief. Experiments were carried out on a real-world dataset from a Chinese consumer finance company. The findings demonstrate that the proposed model outperforms existing benchmarks like LR and RF achieving 91.64% accuracy, 96.89% AUC, and 91.64% KS.

Neagoe et al. (2018) presented a new approach to applying and evaluating Deep CNN versus MLP for financial prediction. They designed a credit scoring model using two neural network classifiers. A MLP with eight layers and a DCNN with thirteen layers. Experiments using the german and australian credit datasets evaluated model performance based on overall accuracy, false alarm rate, and missed alarm rate. The results demonstrated the effectiveness of the proposed approach, as the DCNN significantly outperformed the MLP in both datasets. For the german credit dataset, DCNN achieved an OA of 90.85%, compared to MLP's 81.20%. For the australian credit dataset, DCNN reached an OA of 99.74%, while MLP obtained 90.75%.

4.2.3 Long short-term memory network

Long Short-Term Memory (LSTM) network (Hochreiter and Schmidhuber 1997) is a type of recurrent neural network designed to handle variable length sequences. Comprising LSTM units with input, output, and forget gates, these networks can remember values over extended periods. LSTM is employed to construct layers within neural networks (Yotsawat et al. 2021). The forget gate determines which parts of cell states are worth remembering based on the cell state passed on from the previous time step. While the input gate manages information flow to preserve pertinent data from irrelevant updates (Hochreiter 1998). The cell state gate calculates the new data to be stored in the memory cell and the output gate guides the actual prediction based on the current memory cell. LSTM networks excel at capturing extended dependencies in sequential data. This makes them well-suited for credit risk assessment where historical features are essential (Ala'raj et al. 2022) and they automatically extract relevant features without extensive manual engineering. However, they often require a large amount of data to achieve good performance which can be challenging due to the privacy of the credit data.

Ala'raj et al. (2021) aimed to aid bank management in assessing credit card clients by predicting missed payments. Their model used bidirectional LSTM to calculate the likelihood of missed payments for each customer in the following month. The scores of the model correlated with payment probabilities to enhance consumer credit scoring according to experimental results with 82.40% accuracy, 95.15% specificity, and 78.47% AUC on the non-transactional.

Wang et al. (2018) utilized borrowers' online behavior data to create a consumer credit scoring method employing Attention Mechanism LSTM (AM-LSTM). They treated events as words, transformed them into vectors using the Event2vec model, and employed an attention mechanism LSTM network to predict user default likelihood. The results showed the effectiveness of the proposed model in achieving 71.00% AUC and 31.00% KS on the P2P lending platform dataset.

Ala'raj et al. (2022) employed DL to help bank management in credit card client scoring. They predicted consumer behavior across three dimensions: missed payments, purchasing behavior, and customer grouping based on mathematical expectations of loss. Two models named missed payment prediction LSTM and purchase estimation prediction LSTM were devised to enhance decision-making through customer behavioral grouping. The experiment was tested on the transactional dataset giving the best results with 90.69% accuracy, 72.87% recall, 82.94% KS, and 91.00% AUC.

Adisa et al. (2022) applied LSTM for the financial domain which was rarely used for credit scoring prediction. Their research presented an optimization approach (GA) to determine the optimal parameters for the LSTM model, including epochs, batch size, number of neurons, learning rate, and dropout. The results showed that the optimized LSTM model outperformed both single classifiers and ensemble models with 89.27% accuracy on the australian dataset.

4.2.4 Hybrid deep learning models

Hybrid DL models combine multiple DL techniques or integrate traditional ML methods with DL approaches. Various studies have been proposed to prove the effectiveness of these models.

In a follow-up study, Pławiak et al. (2020) introduced the Deep Genetic Hierarchical Network of Learners (DGHNL) that combined evolutionary computation, EL, and DL. This approach applied to the statlog australian data featured a 16-layer genetic cascade ensemble of classifiers including SVMs, normalization methods, feature extraction, kernel functions, and parameter optimizations. Their proposed model holds potential for use in the banking system achieving 97.39% accuracy on the australian dataset.

Pławiak et al. (2020) introduced the Deep Genetic Hierarchical Network of Learners (DGHNL) that combined various learners, normalization procedures, feature extraction methods, kernel functions, and parameter optimizations. Their approach incorporated DL, EL, genetic feature selection, and optimization, focusing on proper information flow and fusion within the DGHNL structure. The results showed the potential of the proposed model achieving 94.60% accuracy on the german dataset.

Shen et al. (2021) devised a novel DL ensemble model for credit risk assessment by addressing imbalanced credit data. They combined the LSTM network with the AdaBoost algorithm followed by an enhanced SMOTE method for data training. Experimental results indicated the superiority of their proposed model compared to other methods with 80.32% AUC and 39.48% KS on the german credit dataset.

4.3 Ensemble learning models

EL (Dong et al. 2020) is a powerful ML technique that warrants separate examination. This approach involves the integration of multiple learning algorithms to enhance predictive performance by combining the strengths of diverse models. This section describes EL models used for credit scoring.

4.3.1 Gradient boosting decision trees

Gradient Boosting Decision Trees (GBDT) is an ensemble ML method that is widely employed in both classification and regression tasks. The gradient boosting concept involves combining weak base learners often DTs with high bias and low variance to craft a robust and accurate model. GBDT extends the boosting approach by employing a boosting-based error minimization strategy to generate models additively (Liu et al. 2021). The model continues to add trees until a predefined number of trees are built, the loss reaches an acceptable level, or a specified stopping criterion is met. GBDT models can be computationally expensive when dealing with deep trees or complex problems, often consuming a significant amount of memory (Friedman 2001).

Liu et al. (2021) proposed a multi-grained and multi-layered gradient-boosting decision tree (GBDT) for credit scoring. This approach combined the representation learning capability of neural networks with the robustness of ensemble-based methods. Then, they explored the hierarchical representation learning ability of the proposed method. Finally, they enhanced the representation ability of the multi-layered framework by incorporating a

multigrained scanning mechanism. The model achieved favorable results (88.26% accuracy and 94.07% AUC on the australian dataset) by minimizing intra-class distance and increasing inter-class distance.

Zou and Gao (2022) developed a supervised NN-based augmented GBDT-AugBoost-ELM for enhanced credit scoring. This approach utilized bagging ensemble training and boosting ensemble optimization to diversify base learners. The proposed method was tested on several public datasets (table 6) and achieved the best results on the japanese dataset with 86.87% accuracy and 87.91% F1-score.

Bai et al. (2022) proposed a non-parametric ensemble tree model called Gradient Boosting Survival Tree (GBST) to handle heterogeneous industrial data from the chinese consumer financing sector. GBST extended traditional survival tree models by incorporating gradient boosting to optimize survival probabilities over time. This approach effectively minimized total error and proved its superiority in estimating credit risk by achieving 82,51% AUC and 51,64% KS on the 360 finance dataset.

Zhang et al. (2020) introduced a P2P lending online integrated credit scoring model (OICSM) that combined a GBDT with a neural network. OICSM improved credit scoring by handling two types of features and enabling online updates. Tests using real credit datasets from the US and china validated the effectiveness of OICSM achieving the best results by 73.39% AUC on the lending club dataset and 71.76% AUC on the paipaidai dataset. Its advantage in DL and online dynamic update capability contributed to its significant performance improvement.

4.3.2 The extreme gradient boosting

XGBoost model is an ensemble model that combines tree models with gradient boosting. A tree model is a type of supervised model that partitions explanatory variables to best classify the response variable and generates decision trees in parallel (Nobre and Neves 2019). Boosting is a reinforcement algorithm that progressively adds model iterations by adjusting the weights of the weak learners (trees) to minimize error iteration after iteration. Each subsequent tree attempts to reduce the errors introduced by the previous tree. This reduces model bias and improves overall accuracy (Ampountolas et al. 2021). XGBoost not only improves tree models by improving classification performance, but it is also faster than tree model algorithms. In addition, XGBoost is an advanced gradient-boosting model that mitigates overfitting by carefully balancing the reduction of the objective function and the complexity of the model (Ariza-Garzón et al. 2020).

Ampountolas et al. (2021) evaluated ML algorithms on microlending data to classify borrowers into credit categories. They highlighted the success of using customer data and off-the-shelf classifiers like the XGBoost algorithm. The model achieved 88.00% recall, 71.00% specificity, and 78.00% F1-score on the micro-loans dataset. This approach offers a dependable and cost-effective way for developing-world micro-lending institutions to assess creditworthiness without relying on credit history or centralized databases.

Xia et al. (2021) proposed a novel dynamic credit scoring model, SurvXGBoost, which combines survival analysis with the GBDT approach. This model aimed to improve predictability for personal default over time and addresses censoring issues. Compared to benchmark models on a real-world consumer loan dataset, SurvXGBoost achieved an AUC of 68.08%. The results of out-of-sample (68.07% AUC) and out-of-time validation (67.07%

AUC) indicated that SurvXGBoost outperformed the benchmarks in terms of predictability and misclassification cost (64.84%). SurvXGBoost maintained interpretability by providing information on feature importance.

Yotsawat et al. (2021) proposed an improved credit scoring model based on XGBoost classifier using bayesian hyper-parameters optimization (XGBoost-BO). The model involved two steps: data pre-processing to handle missing values and scale the data, followed by bayesian hyper-parameter optimization to tune the XGBoost classifier. The model was evaluated on four public datasets (german, australian, lending club, and polish). Several state-of-the-art classification algorithms were used for predictive comparison. The results showed that the proposed model improved accuracy by 4.10%, 3.03%, and 2.76% on the german, lending club, and australian datasets, respectively. The experimental results confirmed that the XGBoost-BO model was suitable for assessing the creditworthiness of applicants.

4.3.3 Hybrid ensemble learning models

Similar to hybrid machine and DL models, hybrid EL models represent an innovative approach that combines the strengths of different ensemble techniques or integrating them with other ML methods. The development of these models has shown promising results, particularly in enhancing predictive accuracy and robustness in complex datasets.

He et al. (2018) developed a novel ensemble model for credit scoring that addressed the imbalance ratio dataset. Their algorithm modified BalanceCascade for balanced subsets and utilized RF and XGBoost classifiers in a three-stage ensemble. Stacking generated new features from the first layer's outcomes for the second layer optimized via particle swarm optimization. The suggested model demonstrated its superiority in the japanease dataset achieving 88.04% F1-score, 92.79% AUC, 86.22% G-mean, and 75.80% KS.

Rofik et al. (2024) proposed a credit risk assessment model that integrates SMOTE for class imbalance treatment and stacking EL to enhance prediction performance. The base learners in their stacking framework include RF, SVM, Extra Trees Classifier, with XGBoost serving as the meta-learner. The study followed a structured pipeline involving data collection, preprocessing, oversampling, modeling, and evaluation, and used the German Credit dataset with cross-validation. Results demonstrated strong performance with an accuracy of 83.21%, precision of 79.29%, recall of 91.78%, and an F1-score of 85.08%, highlighting the effectiveness of combining SMOTE with stacking techniques for credit scoring tasks.

Zhang et al. (2021) introduced a unique multi-stage ensemble model with outlier adaptation to improve credit scoring. They enhanced the local outlier factor algorithm using bagging for noisy credit data. Their approach also included a novel feature transformation method, stacking-based EL, and self-adaptive parameter optimization. Experiments confirmed the enhanced performance of the model by achieving 93.16% accuracy, 93.45% F1-score, and 96.95% AUC on the japanese dataset.

Jin et al. (2021) introduced a multi-stage ensemble model with a hybrid GA for accurate credit prediction. They addressed data imbalance using the VIHT technique and developed a hybrid GA for selecting features and classifier subsets. The model utilized stacking for final predictions and demonstrated its effectiveness by achieving 93.79% recall and 15.76% F1-score through experiments on the unbalanced polish 2 credit dataset.

Yotsawat et al. (2021) introduced a Cost-sensitive Neural Network Ensemble (CS-NNE) for credit scoring. Their novel approach applied multiple class weights to address imbalanced classes and enhance diversity among base neural networks. The results in Table 6 demonstrated that CS-NNE outperformed single neural networks on real-world imbalanced credit datasets effectively addressing imbalance problems and surpassing existing approaches achieving the highest accuracy (91.30%) on the polish dataset.

Shen et al. (2019) introduced an innovative ensemble model for assessing personal credit risk by integrating SMOTE and classifier optimization. They rebalanced the training dataset with SMOTE by optimizing back propagation neural networks using particle swarm optimization and built an ensemble model combining optimized back propagation neural network classifiers with AdaBoost. The results presented in table 6 showed that the model achieved the best on the australian dataset with 90.58% accuracy, 95.40% F1-score, 91.03% AUC, and 90.94% G-mean.

Jiao et al. (2021) introduced an advanced ensemble model to enhance image feature classification. Their approach combined a CNN for feature extraction and an XGBoost classifier for classification. They optimized the model using an improved particle swarm optimization algorithm to fine-tune hyperparameters. The results on the image and credit dataset in table 6 showed the model's superior performance achieving the best results on the australian dataset with 88.20% accuracy and 87.43% F1-score.

Chen et al. (2020) proposed a unique ensemble model using the generalized shapley value and the choquet integral. They utilized fuzzy measures to capture interactions among base learners using a linear programming model. Representing base learner-predicted values with fuzzy numbers preserved original information. The ensemble model's anticipated value was computed using the Generalized Shapley Choquet Integral (GSCI) aggregation operator. Their GSCI-based ensemble credit scoring approach yielded robust results achieving the best values with 94.53% recall, 90.91% F1-score, and 91.43% AUC on the australian dataset while the best accuracy was 93.35% on the RRDai dataset.

Zhang and Chi (2021) introduced a novel heterogeneous ensemble credit scoring approach to address imbalanced data classification. Their model incorporated LSVM, KNN, MDA, DT, and LR classifiers and adaptively selected the highest AUC base classifiers based on data distribution. Merging these base classifiers yielded predictions that outperformed baseline models by achieving 70.50% AUC on the chilean dataset. It made it useful for actual credit scoring to manage credit risk for financial institutions.

Xing et al. (2024) proposed a stacked ensemble model integrating Random Forest, XGBoost, and TabNet to enhance credit score prediction. By leveraging the strengths of these high-performance models, particularly the DL capabilities of TabNet, the approach addresses limitations of individual classifiers. The ensemble was evaluated using multiple metrics, including Precision, Recall, F1-score, and AUC, and demonstrated superior performance on the Credit Score dataset.

Li et al. (2022) introduced the One-class Classification Driven Dynamical Ensemble Learning (OCDDEL) approach. Unlike using inferred labels, OCDDEL solely relied on accepted applications and their genuine labels. It formed a dynamic ensemble model to handle diverse test applications. By training a one-class classifier, OCDDEL grouped test applications and computed ensemble weights for each case based on similarity with training applicants. The results demonstrated the effectiveness of the proposed model by achieving 89.38% accuracy on the lending club dataset.

Guo et al. (2019) introduced a novel self-adaptive classifier ensemble model utilizing statistics and ML techniques to enhance prediction performance. Their multi-stage approach included data preparation, self-adaptive selection of base classifiers with bayesian optimization-adjusted parameters, and integration of these optimized base classifiers using multi-layer stacking. Testing on real-world credit datasets demonstrated the model's strong performance on the australian dataset with 87.40% accuracy, 86.80% F1-score, and 94.00% AUC.

Tripathi et al. (2018) aimed to merge feature selection and ensemble frameworks. They suggested utilizing feature clustering technique k-means for selection. They then used this reduced dataset with five base classifiers. Aggregating the outputs through weighted voting improved the final prediction. They tested this on three datasets and compared it to existing methods and proved the efficiency of the proposed method by achieving the best results with 87.98% accuracy and 90.69% F1-score on the japanese dataset.

5 Performance evaluation criteria

Selecting appropriate performance evaluation metrics is a critical task in credit scoring, as each metric has its own strengths and limitations (Hand 2009). Certain models may excel under specific criteria but perform poorly under others (Guetari et al. 2023; Gicic and Subasi 2019), making the choice of metric crucial for an accurate assessment.

A widely used approach is the confusion matrix (Table 2), which categorizes predictions into true positives, true negatives, false positives, and false negatives. This foundation supports several derived metrics commonly employed in model evaluation.

Accuracy (formula 3), though intuitive, can be misleading with imbalanced datasets. Precision (formula 4) and recall (formula 5) offer deeper insight into model behavior, particularly in identifying relevant positive cases. The F1-score (formula 6) combines these two into a single metric, especially useful under class imbalance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F1\text{–}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{6}$$

Table 2 Binary confusion matrix

Actual	Predicted	
	Positive	Negative
Positive	*TP*	*FN*
Negative	*FP*	*TN*

Legend: TP = True Positive, FN = False Negative, FP = False Positive, TN = True Negative

The Receiver Operating Characteristic (ROC) curve graphically illustrates the trade-off between sensitivity and specificity across various thresholds. Its summary statistic, the Area Under the Curve (AUC), reflects the overall ability of a model to distinguish between classes.

Additional comprehensive metrics include the Geometric Mean (G-Mean), which balances sensitivity and specificity (formula 7), and the Kolmogorov–Smirnov (KS) statistic, widely used in credit scoring to measure the model's discriminatory power.

$$Specificity = \frac{TN}{TN + FP} \tag{7}$$

Beyond these, several other evaluation metrics have been identified in the literature. Table 3 lists all the evaluation metrics utilized in the reviewed studies.

Table 3 Evaluation metrics and frequency

Evaluation metric	Number of use
Accuracy	49
AUC	31
F1-Score	25
Recall	24
KS	13
Specificity	14
Precision	13
Brier Score	10
G-Mean	10
Type II error	7
H-measure	6
Type I error	5
Log loss	3
Matthews Correlation Coefficient	3
Misclassification Cost	2
ROC area	2
Balanced Accuracy	1
Bookmaker Informedness	1
C-index	1
Detection Rate	1
False Alarm Rate	1
Mean Absolute Error	1
Missed Alarm Rate	1
Partial Gini index	1
Root Mean Squared Erro	1
The Average Efficiency	1
The Overall Efficiency	1
The Relative Absolute Error	1

6 Results

6.1 Study selection and PRISMA flow diagram

A total of 345 studies were identified through database searches and backward snowballing. After removing duplicates and applying inclusion criteria, 63 primary studies published between 2018 and 2024 were included in the final synthesis (Fig. 1). These studies comprise 48 journal articles (79%) and 13 conference papers (21%) and reflect a broad global contribution from both academic and industrial institutions.

6.2 Study characteristics and summary tables

The 63 included studies were categorized into three main groups: traditional ML, DL, and EL, including hybrid approaches within each category. The most frequently used public datasets were German, Australian, and Japanese credit datasets, while about one-third of studies relied on proprietary institutional datasets.

Tables 4, 5, and 6 summarize all studies, highlighting the models, datasets, and evaluation metrics, providing a foundation for comparative analysis in the next section.

6.3 Performance analysis of reviewed studies

A comparative synthesis of performance metrics was conducted across the reviewed studies. Among these metrics, accuracy and AUC are the most frequently reported together across the reviewed studies. This is because accuracy provides an overall correctness measure,

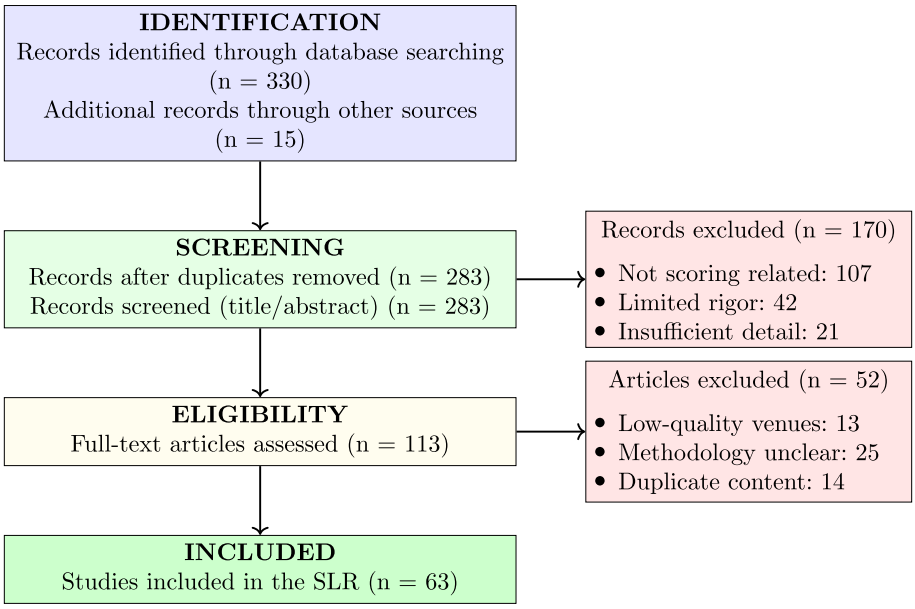

Fig. 1 PRISMA 2020 Flow Diagram for SLR

Table 4 Summary studies of machine learning models

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Aji and Dhini (2019)	Indonesia Bank	RF+AdaBoost	72.95%	73.00%	NA	70.40%	NA	NA	NA	NA
Zhang et al. (2018)	Australian	NCSM	91.71%	NA	NA	NA	NA	NA	NA	NA
	German	NCSM	82.14%	NA	NA	NA	NA	NA	NA	NA
Yao and Chen (2019)	German	SVM-RF	83.55%	54.91%	NA	90.46%	NA	84.00%	NA	NA
	Australian	SVM-RF	87.94%	83.85%	NA	91.69%	NA	92.10%	NA	NA
Dumitrescu et al. (2022)	Australian	PLTR	NA	NA	NA	NA	NA	92.99%	NA	74.25%
	Taiwan	PLTR	NA	NA	NA	NA	NA	77.80%	NA	42.57%
	Housing	PLTR	NA	NA	NA	NA	NA	90.11%	NA	73.41%
Teles et al. (2021)	Bank Institution	SVM	98.34%	91.00%	100%	NA	96.00%	NA	NA	NA
Tran et al. (2021)	Kalapa	RF	NA	NA	NA	NA	83.00%	81.00%	NA	NA
Trivedi (2020)	German	RF+CS	93.12%	NA	NA	NA	93.10%	NA	NA	NA
Parvin and Saleena (2020)	Australian	RF	88.41%	80.00%	87.00%	NA	84.00%	NA	NA	NA
Cao et al. (2021)	Lending Club	LR	86.58%	NA	NA	NA	NA	NA	NA	NA
Moscato (2021)	Lending Club	RF-RUS	64.00%	63.00%	NA	68.00%	NA	71.70%	65.60%	NA
Tripathi et al. (2019)	Australian	Proposed	92.69%	97.16%	NA	88.46%	NA	NA	92.68%	NA
	Janapense	Proposed	89.06%	96.16%	NA	83.81%	NA	NA	89.74%	NA
	German-categorical	Proposed	85.18%	93.29%	NA	65.06%	NA	NA	77.89%	NA
	German-numerical	Proposed	84.69%	94.52%	NA	65.83%	NA	NA	78.86%	NA
Goh et al. (2020)	German	HS-RF	76.40%	NA	NA	NA	84.53%	80.44%	NA	NA
	Australia	HS-RF	87.38%	NA	NA	NA	93.66%	86.14%	NA	NA
	Lending Club	HS-RF	85.71%	NA	NA	NA	86.74%	85.71%	NA	NA
Li et al. (2021)	German	gcForest	81.20%	NA	NA	NA	NA	86.80%	NA	NA
	Australian	gcForest	88.55%	NA	NA	NA	NA	94.25%	NA	NA
	Japanese	gcForest	88.99%	NA	NA	NA	NA	96.02%	NA	NA
Nalic et al. (2020)	Bosnia	GLM+DT	87.69%	57.22%	NA	NA	87.69%	NA	NA	NA
Ariza-Garzón et al. (2020)	Lending Club	LR	78.10%	NA	NA	NA	NA	66.60%	NA	23.90%

Table 4 (continued)

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Zhang et al. (2021)	Australian	Proposed	90.58%	NA	NA	NA	88.88%	94.83%	NA	52.26%
	German	Proposed	79.50%	NA	NA	NA	82.59%	83.84%	NA	46.27%
	Japanese	Proposed	89.85%	NA	NA	NA	90.14%	95.30%	NA	52.15%
	Taiwan	Proposed	80.50%	NA	NA	NA	48.47%	79.37%	NA	13.50%
	Creator	Proposed	99.77%	NA	NA	NA	99.71%	99.84%	NA	52.23%
Boughaci et al. (2021)	Australian	K-means+RF	NA	96.50%	96.50%	NA	96.50%	NA	NA	NA
	German	K-means+RF	NA	90.60%	90.90%	NA	90.40%	NA	NA	NA
	Japanese	K-means+RF	NA	97.80%	97.80%	NA	97.80%	NA	NA	NA
	Taiwan	K-means+RF	NA	100.00%	100.00%	NA	100.00%	NA	NA	NA
	Polish	K-means+RF	NA	99.60%	99.60%	NA	99.60%	NA	NA	NA
	Indian	K-means+RF	NA	98.80%	98.80%	NA	98.80%	NA	NA	NA
Bao et al. (2019)	German	Proposed	80.80%	72.10%	63.50%	NA	NA	NA	NA	NA
	Chinese P2P credit	Proposed	92.00%	69.30%	80.90%	NA	NA	NA	NA	NA
Ibrahim and Olagunju (2022)	Bank of Agriculture	SOM+CART	96.70%	97.80%	NA	95.70%	NA	NA	NA	NA
Suleiman et al. (2021)	Bank of Agriculture	SOM+KNN	96.30%	97.80%	NA	95.10%	NA	NA	NA	NA
		SOM+NN	97.30%	97.80%	NA	97.10%	NA	NA	NA	NA
Yuan et al. (2022)	Australian	K-means+SVDD	NA	NA	NA	NA	NA	84.86%	85.58%	NA
	German	K-means+SVDD	NA	NA	NA	NA	NA	65.71%	62.36%	NA
	Japanese	K-means+SVDD	NA	NA	NA	NA	NA	85.17%	85.11%	NA
	Chinese Companies	K-means+SVDD	NA	NA	NA	NA	NA	86.33%	86.12%	NA
Jin et al. (2021)	Australian	Proposed	NA	NA	NA	NA	NA	93.98%	86.67%	77.84%
	German	Proposed	NA	NA	NA	NA	NA	77.96%	69.97%	44.84%
	Japanese	Proposed	NA	NA	NA	NA	NA	92.45%	86.64%	76.47%
	AER	Proposed	NA	NA	NA	NA	NA	95.02%	87.27%	77.76%
Dm and Mm (2018)	Equity Bank	SVM	86.12%	94.55%	78.31%	NA	85.67%	NA	NA	NA
Wang and Li (2019)	RenRenDai	SVM	NA	NA	93.00%	NA	NA	NA	NA	NA
Syed Nor et al. (2019)	Debt Agency	DT	83.29%	99.00%	NA	06.62%	NA	NA	NA	NA
Khedr et al. (2021)	ABE Dataset	DT	94.85%	96,81%	96,70%	NA	96.75%	NA	NA	NA

Table 4 (continued)

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Maharjan (2022)	German Loan	DT	86,00%	96.03%	85.53%	NA	90.48%	NA	NA	NA
Mukid et al. (2018)	Bank from Indonesia	KNN	82.4%	99.34%	NA	11.11%	NA	NA	NA	NA
Loo et al. (2023)	Kaggle	KNN	89.00%	89.00%	89.00%	NA	89.00%	NA	NA	NA
Pratiwi et al. (2019)	Bank from Indonesia	KNN	98.11%	NA	NA	NA	NA	NA	NA	NA

Table 5 Summary studies of deep learning models

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Kazemi et al. (2021)	Australian	GA + NN	97.78%	NA	NA	NA	NA	NA	NA	NA
	German	GA + NN	87.10%	NA	NA	NA	NA	NA	NA	NA
Kazemi et al. (2023)	German	GA + NN	67.49%	NA	NA	NA	NA	75.92%	NA	NA
	Australian	GA + NN	91.91%	NA	NA	NA	NA	92.60%	NA	NA
Ala'raj et al. (2021)	Non Transactional	LSTM	82.40%	37.51%	NA	95.15%	NA	78.47%	NA	43.00%
Dastile and Celik (2021)	German	2D CNN	82.00%	NA	NA	NA	NA	NA	NA	NA
	Australian	2D CNN	95.00%	NA	NA	NA	NA	NA	NA	NA
	HMEQ	2D CNN	82.00%	NA	NA	NA	NA	NA	NA	NA
Wang et al. (2018)	P2P lending platform	AM-LSTM	NA	NA	NA	NA	NA	71.00%	NA	31.00%
Zhu et al. (2018)	Chinese Company	Relif-CNN	91.64%	NA	NA	NA	NA	96.89%	NA	91.64%
Ala'raj et al. (2022)	Transactional	MP-LSTM	90.69%	72.87%	NA	96.25%	82.94%	91.00%	NA	71.00%
Diaconescu and Neagoe (2020)	German	DLNN	84.83%	NA	NA	NA	NA	NA	NA	NA
Pławiak et al. (2019)	Australian	DGCEC	97.39%	NA	NA	NA	NA	NA	NA	NA
Pławiak et al. (2020)	German	DGHNL	94.60%	NA	NA	NA	NA	NA	NA	NA
Shen et al. (2021)	German Credit	LSTM	NA	NA	NA	NA	NA	80.32%	NA	39.48%
	Taiwan Credit	LSTM	NA	NA	NA	NA	NA	74.90%	NA	39.48%
Neagoe et al. (2018)	German Credit	DCNN	90.85%	NA	NA	NA	NA	NA	NA	NA
	Australian Credit	DCNN	99.74%	NA	NA	NA	NA	NA	NA	NA
Adisa et al. (2022)	Australian Credit	LSTM	89.27%	NA	NA	NA	NA	NA	NA	NA

while AUC reflects the model's ability to distinguish between classes, which is critical for imbalanced credit scoring datasets.

To further understand the comparative performance of different ML models, Tables 7 to 10 present the accuracy and AUC results reported in selected studies using the most common benchmark datasets: German, Australian, Japanese, and Lending Club datasets.

Across the German dataset (Table 7), the GA + NN model (Kazemi et al. 2023) achieved the highest accuracy (91.91%) and AUC (92.60%), followed by ensemble approaches such as AugBoost-ELM (Zou and Gao 2022) and gcForest (Li et al. 2021). On the Australian and Japanese datasets (Tables 8, 9), Zhang et al.'s multi-stage ensemble (Zhang et al. 2021) consistently outperformed other methods, achieving accuracies above 91% and AUCs above 96%. For the Lending Club dataset (Table 10), the GSCI model (Chen et al. 2020) led performance with 91.70% accuracy and 93.78% AUC.

Overall, hybrid and EL approaches, particularly those integrating neural networks, optimization algorithms, or boosting, demonstrated consistently superior performance. Traditional models like logistic regression or standard SVMs, while occasionally competitive, generally exhibited lower discrimination, highlighting the advantage of advanced hybrid architectures in credit scoring under imbalanced or noisy datasets.

6.3.1 Visual comparison of model performance

To complement the tabular summary of model performance, Figs. 2, 3, and 4 illustrate the comparative accuracy (± standard deviation) of ML, DL, and EL models in credit scoring. These plots provide a visual representation of variability across studies and highlight the models that consistently achieve high predictive performance.

Figures 2, 3 and 4 show that all three families of models achieve relatively high predictive accuracy, with most methods consistently above 80%. ML models demonstrate strong performance, with Random Forest and the proposed hybrid ML approaches achieving the highest accuracies. DL models, such as CNN and hybrid DL configurations, provide robust performance with lower variability, indicating their ability to handle complex, high-dimensional credit data. EL models, particularly XGB-BO and the proposed ensemble, capture peak performance in certain studies, although with slightly higher variability. Overall, these visualizations support the tabular findings, emphasizing that hybrid and ensemble architectures offer top performance, while DL ensures stable outcomes and classical ML models remain reliable benchmarks.

6.4 Theoretical contributions from scientific mapping

To analyze the intellectual structure of the credit scoring literature, we applied science mapping methods using VOSviewer (Van Eck and Waltman 2010), conducting two complementary analyses: bibliographic coupling and keyword co-occurrence.

6.4.1 Bibliographic coupling

Bibliographic coupling identified three main conceptual clusters among the reviewed studies:

Table 6 Summary studies of ensemble learning models

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Liu et al. (2021)	Australian	mg-mGBDT	88.26%	NA	NA	NA	NA	94.07%	NA	NA
	Lending Club	mg-mGBDT	67.86%	NA	NA	NA	NA	73.74%	NA	NA
	Taiwan	mg-mGBDT	69.22%	NA	NA	NA	NA	74.90%	NA	NA
	WE.com	mg-mGBDT	85.11%	NA	NA	NA	NA	90.39%	NA	NA
	Japanese	mg-mGBDT	86.90%	NA	NA	NA	NA	93.11%	NA	NA
	German	mg-mGBDT	76.53%	NA	NA	NA	NA	78.46%	NA	NA
Zou and Gao (2022)	Australian	AugBoost-ELM	84.39%	84.85%	84.62%	NA	84.62%	94.22%	NA	NA
	German	AugBoost-ELM	76.17%	92.45%	77.75%	NA	78.61%	94.22%	NA	NA
	Japanese	AugBoost-ELM	86.87%	85.82%	90.10%	NA	87.91%	93.99%	NA	NA
	Taiwan	AugBoost-ELM	61.94%	73.22%	90.10%	NA	67.11%	75.16%	NA	NA
Bai et al. (2022)	360 Finance	GBST	NA	NA	NA	NA	NA	82.51%	NA	51.64%
	Lending Club loan	GBST	NA	NA	NA	NA	NA	77.02%	NA	42.45%
Ampountolas et al. (2021)	Micro-Loans	XGBoost	NA	88.00%	NA	71.00%	78.00%	NA	NA	NA
He et al. (2018)	Australian	Proposed	NA	NA	NA	NA	85.41%	92.79%	86.69%	76.75%
	Japanese	Proposed	NA	NA	NA	NA	88.04%	92.79%	86.22%	75.80%
	German	Proposed	NA	NA	NA	NA	84.96%	79.20%	60.75%	46.43%
	DefaultData	Proposed	NA	NA	NA	NA	47.20%	76.67%	59.03%	41.29%
	PPDaiData	Proposed	NA	NA	NA	NA	05.99%	69.10%	17.72%	27.42%
	LC2017Q1Data	Proposed	NA	NA	NA	NA	99.35%	66.10%	00.00%	24.68%
Zhang et al. (2020)	Lending Club	OICSM	NA	NA	NA	NA	NA	73.39%	NA	NA
	Paipaidai	OICSM	NA	NA	NA	NA	NA	71.76%	NA	NA
Guo et al. (2019)	Australian	Proposed	87.40%	NA	NA	NA	86.80%	94.00%	NA	NA
	German	Proposed	78.30%	NA	NA	NA	85.00%	80.60%	NA	NA
	Japanese	Proposed	87.00%	NA	NA	NA	83.70%	94.20%	NA	NA
Jin et al. (2021)	German	Proposed	NA	82.67%	NA	NA	59.99%	NA	70.31%	NA
	Polish 1	Proposed	NA	87.98%	NA	NA	13.66%	NA	63.11%	NA
	Polish 2	Proposed	NA	93.79%	NA	NA	15.76%	NA	64.71%	NA

Table 6 (continued)

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
Yotsawat et al. (2021)	German	XGB-BO	79.50%	90.14%	NA	54.67%	NA	80.50%	69.95%	NA
	Australian	XGB-BO	88.70%	89.29%	NA	87.96%	NA	93.25%	88.59%	NA
	Lending Club	XGB-BO	67.86%	73.41%	NA	62.32%	NA	72.48%	67.49%	NA
	Polish	XGB-BO	98.11%	99.78%	NA	56.48%	NA	95.32%	74.61%	NA
Yotsawat et al. (2021)	German	CS-NNE	74.40%	NA	NA	NA	NA	80.11%	NA	NA
	Polish	CS-NNE	91.30%	NA	NA	NA	NA	88.62%	NA	NA
	Australian	CS-NNE	84.93%	NA	NA	NA	NA	91.31%	NA	NA
	Lending Club	CS-NNE	63.61%	NA	NA	NA	NA	70.82%	NA	NA
Shen et al. (2019)	German	Proposed	78.70%	NA	NA	NA	66.35%	81.02%	75.69%	NA
	Australian	Proposed	90.58%	NA	NA	NA	95.40%	91.03%	90.94%	NA
Zhang et al. (2021)	Australian	Proposed	92.36%	NA	NA	NA	91.80%	96.65%	NA	NA
	German	Proposed	79.50%	NA	NA	NA	85.09%	83.12%	NA	NA
	Japanese	Proposed	93.16%	NA	NA	NA	93.45%	96.95%	NA	NA
	Taiwan	Proposed	82.93%	NA	NA	NA	50.99%	80.91%	NA	NA
Zhang and Chi (2021)	German	Proposed	NA	NA	NA	NA	NA	68.40%	NA	NA
	DefaultData	Proposed	NA	NA	NA	NA	NA	66.10%	NA	NA
	Chilean	Proposed	NA	NA	NA	NA	NA	70.50%	NA	NA
	GMSC	Proposed	NA	NA	NA	NA	NA	58.90%	NA	NA
Chen et al. (2020)	Australian	GSCI	91.16%	94.53%	NA	91.94%	90.91%	91.43%	NA	NA
	German	GSCI	77.75%	67.12%	NA	74.87%	58.22%	70.42%	NA	NA
	Japanese	GSCI	89.13%	92.20%	NA	86.58%	87.84%	89.48%	NA	NA
	Taiwan China	GSCI	82.72%	60.75%	NA	79.38%	47.29%	65.98%	NA	NA
	RRDai	GSCI	93.35%	64.76%	NA	66.02%	73.12%	82.64%	NA	NA
	Prosper Loan	GSCI	89.97%	95.00%	NA	83.28%	94.65%	55.27%	NA	NA
	Lending Club	GSCI	91.70%	98.24%	NA	83.26%	87.44%	93.78%	NA	NA
Li et al. (2022)	Lending Club	OCDDEL	89.38%	NA	NA	NA	NA	NA	NA	46.06%
	Rong360	OCDDEL	81.03%	NA	NA	NA	NA	NA	NA	48.27%
Jiao et al. (2021)	German	APSO-XGBoost	77.48%	NA	NA	NA	77.96%	NA	NA	NA

Table 6 (continued)

Studies	Dataset	Model	Accuracy	Recall	Precision	Specificity	F1 Score	AUC	G-Mean	KS
	Australian	APSO-XGBoost	88.20%	NA	NA	NA	87.43%	NA	NA	NA
	P2P-LC	APSO-XGBoost	67.63%	NA	NA	NA	66.01%	NA	NA	NA
	P2P-We	APSO-XGBoost	84.72%	NA	NA	NA	87.08%	NA	NA	NA
Tripathi et al. (2018)	Australian	Proposed	87.32%	NA	NA	NA	88.93%	NA	NA	NA
	German	Proposed	77.12%	NA	NA	NA	85.42%	NA	NA	NA
	Japanese	Proposed	87.98%	NA	NA	NA	90.69%	NA	NA	NA
Xia et al. (2021)	Loan Transactions	SurvXGBoost	NA	NA	NA	NA	NA	68.08%	NA	NA
Rofik et al. (2024)	German	Stacking	83.21%	91.78%	79.29%	NA	85.08%	NA	NA	NA
Xing et al. (2024)	Credit score	Proposed	NA	79.66%	79.93%	NA	79.68%	91.72%	NA	NA

Table 7 Comparative evaluation of models using the German dataset based on Accuracy and AUC, ranked by combined performance

Rank	Study	Model	Accuracy	AUC
1	Kazemi et al. (2023)	GA + NN	91.91%	92.60%
2	Zou and Gao (2022)	AugBoost-ELM	76.17%	94.22%
3	Li et al. (2021)	gcForest	81.20%	86.80%
4	Yao and Chen (2019)	SVM-RF	83.55%	84.00%
5	Zhang et al. (2021)	Proposed	79.50%	83.84%
6	Zhang et al. (2021)	Multi-stage ensemble	79.50%	83.12%
7	Shen et al. (2019)	Proposed	78.70%	81.02%
8	Yotsawat et al. (2021)	XGB-BO	79.50%	80.50%
9	Guo et al. (2019)	Proposed	78.30%	80.60%
10	Goh et al. (2020)	HS-RF	76.40%	80.44%
11	Yotsawat et al. (2021)	CS-NNE	74.40%	80.11%
12	Liu et al. (2021)	mg-mGBDT	76.53%	78.46%
13	Chen et al. (2020)	GSCI	77.75%	70.42%

Table 8 Ranked comparative evaluation of models using the Australian dataset based on Accuracy and AUC

Rank	Study	Model	Accuracy (%)	AUC (%)
1	Zhang et al. (2021)	Multi-stage ensemble	92.36	96.65
2	Kazemi et al. (2023)	GA + NN	91.91	92.60
3	Zhang et al. (2021)	Proposed	90.58	94.83
4	Shen et al. (2019)	Proposed	90.58	91.03
5	Li et al. (2021)	gcForest	88.55	94.25
6	Yotsawat et al. (2021)	XGB-BO	88.70	93.25
7	Liu et al. (2021)	mg-mGBDT	88.26	94.07
8	Guo et al. (2019)	Proposed	87.40	94.00
9	Yao and Chen (2019)	SVM-RF	87.94	92.10
10	Goh et al. (2020)	HS-RF	87.38	86.14
11	Zou and Gao (2022)	AugBoost-ELM	84.39	94.22
12	Chen et al. (2020)	GSCI	91.16	91.43
13	Yotsawat et al. (2021)	CS-NNE	84.93	91.31

Table 9 Ranked comparative evaluation of models using the Japanese dataset based on Accuracy and AUC

Rank	Study	Model	Accuracy (%)	AUC (%)
1	Zhang et al. (2021)	Multi-stage ensemble	93.16	96.95
2	Li et al. (2021)	gcForest	88.99	96.02
3	Zhang et al. (2021)	Proposed	89.85	95.30
4	Guo et al. (2019)	Proposed	87.00	94.20
5	Zou and Gao (2022)	AugBoost-ELM	86.87	93.99
6	Liu et al. (2021)	mg-mGBDT	86.90	93.11
7	Chen et al. (2020)	GSCI	89.13	89.48

Table 10 Ranked comparative evaluation of models using the Lending Club dataset based on Accuracy and AUC

Rank	Study	Model	Accuracy (%)	AUC (%)
1	Chen et al. (2020)	GSCI	91.70	93.78
2	Goh et al. (2020)	HS-RF	85.71	85.71
3	Ariza-Garzón et al. (2020)	LR	78.10	66.60
4	Liu et al. (2021)	mg-mGBDT	67.86	73.74
5	Yotsawat et al. (2021)	XGB-BO	67.86	72.48
6	Moscato (2021)	RF-RUS	64.00	71.70
7	Yotsawat et al. (2021)	CS-NNE	63.61	70.82

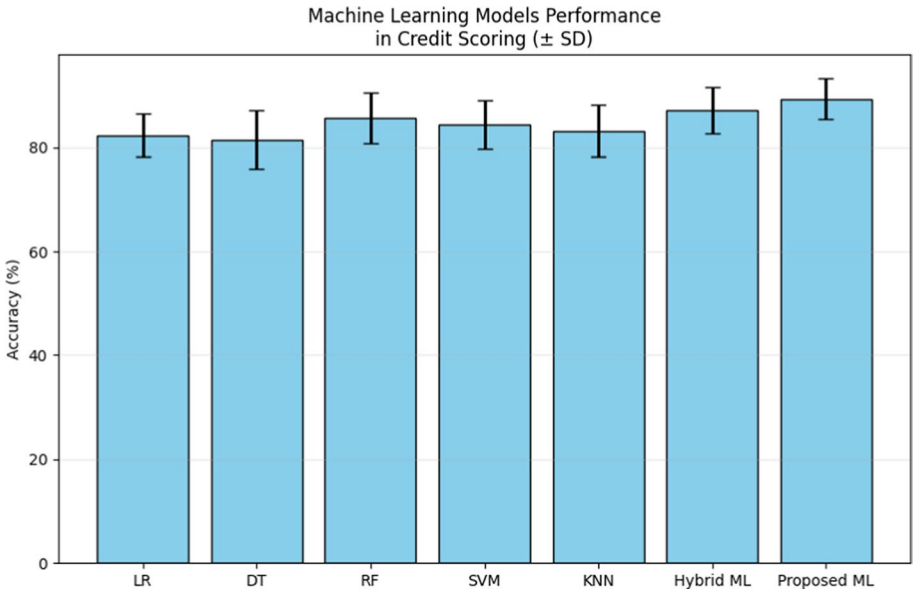

Fig. 2 Comparative accuracy (± standard deviation) of machine learning models in credit scoring

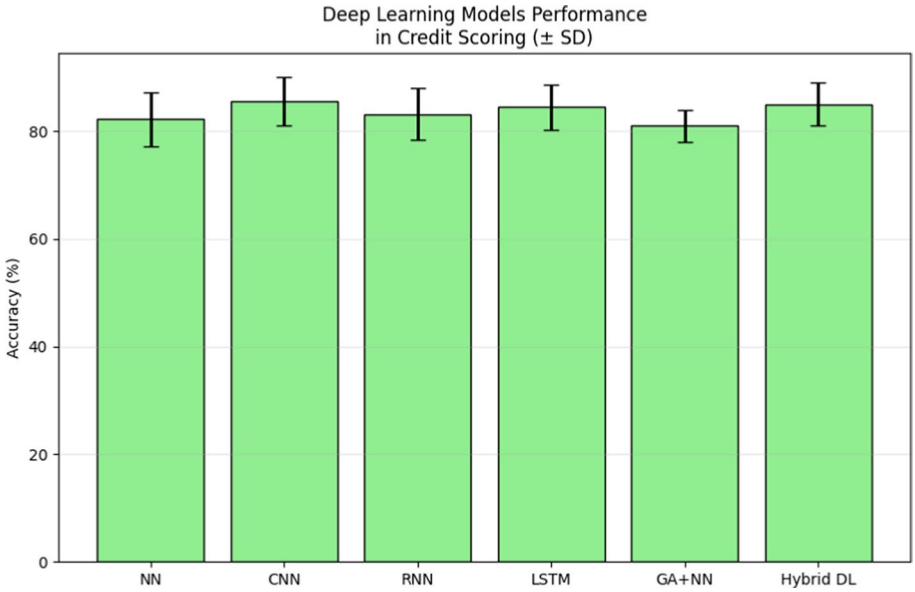

Fig. 3 Comparative accuracy (± standard deviation) of DL models in credit scoring

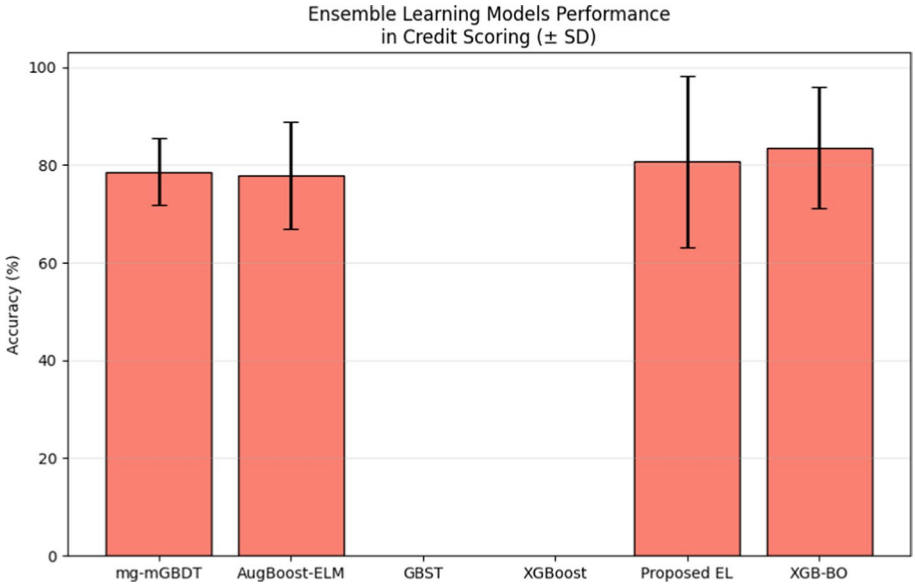

Fig. 4 Comparative accuracy (± standard deviation) of EL models in credit scoring

- **Cluster 1:** Traditional statistical approaches and incremental improvements.
- **Cluster 2:** Ensemble-based ML techniques emphasizing predictive performance.
- **Cluster 3:** Emerging DL applications and hybrid modeling frameworks.

This structure illustrates a progression from classical models to more advanced and automated learning systems. Figure 5 visualizes the bibliographic coupling network, highlighting the thematic organization of the field.

6.4.2 Keyword co-occurrence

Keyword co-occurrence analysis revealed six thematic clusters, reflecting both established and emerging research areas:

- **Cluster 1:** classification models, credit scoring, ML, neural networks
- **Cluster 2:** credit analysis, gradient methods, multi-model approaches
- **Cluster 3:** explainable AI, credit score prediction, model interpretability
- **Cluster 4:** ensemble models, novel applications, multi-stage frameworks
- **Cluster 5:** DL, predictive modeling
- **Cluster 6:** feature selection and engineering

These clusters highlight key research trends, including the adoption of hybrid and EL methods, DL techniques, explainable AI, and feature engineering. Figure 6 presents the corresponding keyword co-occurrence map.

6.5 Citation analysis

Table 11 summarizes the five most cited articles globally, highlighting their contributions to credit scoring research.

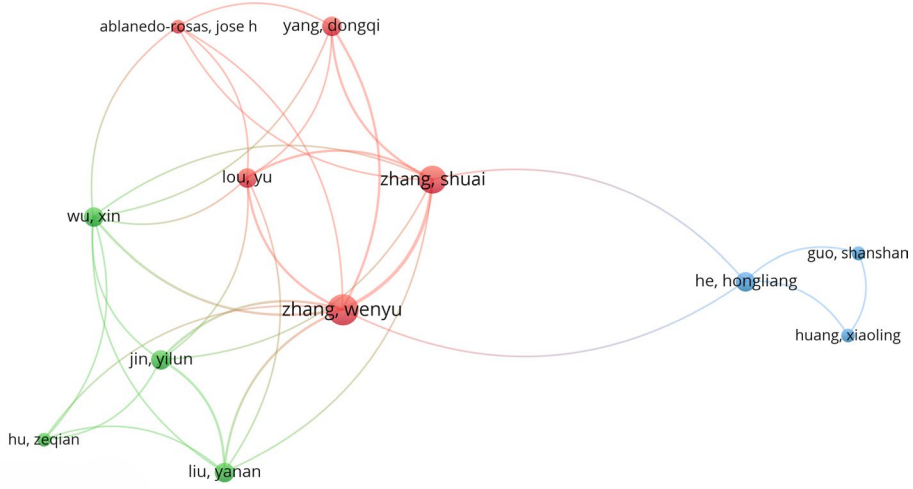

Fig. 5 Bibliographic coupling network of included studies

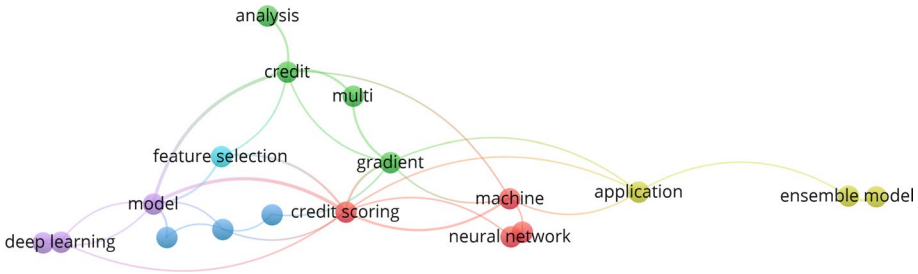

Fig. 6 Keyword co-occurrence map showing thematic clusters

Table 11 Global most cited documents performance

Rank	Study	Cites	title
1	Dumitrescu et al. (2022)	410	ML for credit scoring: Improving logistic regression with non-linear decision-tree effects
2	He et al. (2018)	313	A novel ensemble method for credit scoring: Adaption of different imbalance ratios
3	Moscato (2021)	284	A benchmark of ML approaches for credit score prediction
4	Shen et al. (2021)	225	A new DL ensemble credit risk evaluation model with an improved synthetic minority oversampling technique
5	Bao et al. (2019)	222	Integration of unsupervised and supervised ML algorithms for credit risk assessment

Dumitrescu et al. (2022) introduced the penalised logistic tree regression (PLTR), combining decision tree rules with logistic regression to enhance accuracy while maintaining interpretability. He et al. (2018) proposed an ensemble model that adapts to varying class imbalance ratios using BalanceCascade, random forest, XGBoost, stacking, and particle swarm optimization, influencing subsequent studies addressing data imbalance. Moscato (2021) conducted a benchmarking study of widely used ML models for peer-to-peer lending, assessing both predictive performance and interpretability through explainable AI. Shen et al. (2021) combined an improved SMOTE method with LSTM and AdaBoost in a DL ensemble framework, achieving strong performance on imbalanced datasets. Bao et al. (2019) demonstrated that integrating unsupervised learning at the consensus and clustering stages with supervised models significantly improves performance, underscoring the value of hybrid learning strategies in credit risk assessment.

6.6 Variables and feature selection

The reviewed studies employed a wide range of input variables, which we grouped into four main categories, reflecting the different aspects of borrower information:

- *Demographic variables:* age, sex, marital status, employment type, number of dependents, and residence type (such as own, rent). These variables describe the personal back-

ground of the borrower.

- *Financial variables:* annual income, loan amount, debt-to-income ratio, monthly loan payments, number of existing credit lines, and account balances. These indicate the borrower's current financial status and obligations.
- *Behavioral variables:* payment history, delinquency counts, credit utilization ratio, previous defaults, and loan repayment behavior. These variables capture the borrower's past creditworthiness and financial behavior (Bhandary et al. 2023).
- *Transaction-level variables:* recent credit inquiries, number of open accounts, new credit lines opened, and overdraft history, representing recent financial activity and credit engagement.

These categories encompass the most commonly used variables in popular credit scoring datasets such as the German, Australian, and Japanese datasets, which formed a significant part of the reviewed literature.

Feature selection techniques were frequently employed to enhance model performance by identifying the most relevant variables and reducing noise. Commonly used methods include:

- Information Gain (Lenka et al. 2022; Trivedi 2020)
- Gain Ratio (Trivedi 2020)
- Chi-square (Trivedi 2020)
- PCA (Lenka et al. 2022)
- GA (Lenka et al. 2022)
- Attribute Weighting (Cao et al. 2021)
- HS integrated for feature selection (Goh et al. 2020)
- Relief-based feature selection (Zhu et al. 2018)
- K-means clustering applied as a feature clustering technique for selection (Tripathi et al. 2018)

These techniques serve to improve input quality by selecting or transforming features, ultimately leading to better predictive accuracy in credit scoring models.

7 Discussion

This section discusses the findings of the systematic review in relation to the five research questions defined in the methodology. It synthesizes the evidence regarding the most commonly used ML models, their strengths and limitations, evaluation practices, emerging trends, and key challenges in credit scoring. In addition to addressing these questions, the section also presents a comparative analysis of model performance across studies to highlight which approaches have demonstrated superior predictive capabilities in recent literature.

7.1 RQ1: What are the most widely used ML models for credit scoring?

The most frequently applied ML models in credit scoring are detailed in Sect. 4. Tables 4, 5, and 6 provide an overview of conventional ML, DL, and ensemble models, along with the datasets and evaluation metrics reported across the reviewed studies.

Analysis of the reviewed studies indicates that hybrid ensemble models are the most widely used, reflecting their ability to combine multiple algorithms, leverage complementary strengths, improve predictive accuracy, and manage heterogeneous borrower profiles. Conventional ML models, such as LR, DTs, and SVM, remain relevant, particularly when interpretability and simplicity are prioritized. DL models, while promising for large-scale or high-dimensional datasets, are less frequently adopted due to their data-intensive nature and limited interpretability.

Overall, the evidence suggests that practitioners prioritize hybrid ensemble approaches when predictive performance is critical, whereas conventional ML models are preferred when transparency or smaller datasets are considered.

7.2 RQ2: What are the strengths and limitations of ML models used for credit scoring?

ML models exhibit various strengths and limitations when applied to credit scoring. Understanding these factors is essential for assessing their suitability in financial decision-making. The following subsections detail the key strengths and weaknesses identified in the literature.

7.2.1 Strengths of machine learning models

A major strength of ML models is their ability to generate accurate predictions by learning patterns in historical data that reflect applicant behavior over time.

Many ML workflows include automated feature selection methods, reducing the need for manual engineering and potentially uncovering novel predictive features. This automation enhances the efficiency and effectiveness of the credit scoring process. For example, DTs are particularly effective for small datasets, providing both reliable predictions and interpretability, which is essential for regulatory compliance.

DL models are well-suited for large datasets due to their ability to automatically capture complex, non-linear relationships that traditional models often fail to identify. Their ability to handle high-dimensional data enables capturing subtle interactions among features, which can significantly improve predictive performance in scenarios with complex borrower profiles.

EL models enhance predictive performance in credit scoring by aggregating the outputs of multiple base models, thereby reducing both bias and variance, especially when handling heterogeneous borrower profiles. Evidence from the reviewed studies shows that ensemble methods, particularly random forests and XGBoost, consistently achieve higher accuracy and AUC scores than single classifiers, indicating their widespread preference in practical credit scoring applications.

Hybrid models are another effective approach, integrating multiple algorithms to capture diverse patterns and relationships in the data, thereby improving predictive accuracy and

robustness. They can also be customized for specific datasets and problems by selecting the most suitable algorithms, making them highly adaptable to different credit scoring scenarios. The flexibility of hybrid models allows financial institutions to tailor models to specific borrower segments, supporting more nuanced risk assessment strategies.

7.2.2 Limitations of machine learning models

Despite their strengths, ML models exhibit several limitations, particularly in credit scoring applications. One major challenge is their reliance on high-quality labeled data, which is often difficult to obtain due to confidentiality and privacy concerns. Inadequate feature extraction and low-quality labeled data can significantly degrade the performance of conventional learning models, as illustrated by the limited effectiveness of DL models with simpler architectures.

Model interpretability is another critical issue, particularly with complex ML techniques such as DL. Despite their impressive accuracy, these models are often regarded as black boxes, making their decisions difficult to interpret and raising transparency concerns. This challenge complicates the ability of lenders and borrowers to understand the model's logic. Moreover, DL models are highly sensitive to the quality and quantity of training data which makes them susceptible to bias, especially with imbalanced datasets. While techniques like SMOTE address data imbalance, they require careful handling to avoid overfitting and ensure robust performance.

The limited adoption of DL models in credit scoring may stem from their complexity and lack of interpretability, especially compared to traditional and ensemble methods. The fewer articles focusing on DL for credit scoring than other ML techniques underscore this limitation. While ensemble models can enhance the performance of credit scoring models, they may reduce interpretability compared to single models. The complexity of interpreting ensemble models with multiple classifiers can be a significant drawback.

Similarly, integrating multiple models in hybrid techniques increases the complexity of model construction, tuning, and validation. This added complexity often results in higher computational overhead and extended development times. Moreover, hybrid models demand substantial computational resources due to the integration of multiple algorithms. Such resource intensity may hinder the scalability, practicality, and interpretability of hybrid techniques, particularly for smaller institutions with constrained computational capabilities. Hybrid models may also suffer from overfitting, particularly when combining complex or deeply nested architectures.

7.2.3 Synthesis and practical implications

Overall, the literature indicates that ensemble and hybrid ML models are the most widely used and effective approaches for credit scoring, largely due to their superior predictive performance and ability to handle heterogeneous datasets. DL models offer high potential for large, high-dimensional datasets but are constrained by interpretability and data availability issues. For practitioners, this suggests a strategic approach: use ensemble or hybrid models where predictive performance is critical and data availability is sufficient, while balancing interpretability requirements to satisfy regulatory and operational needs.

From a research perspective, these findings highlight the need to develop methods that maintain the accuracy of complex models while improving interpretability and robustness. Techniques such as explainable AI and careful data preprocessing are practical solutions that bridge the gap between predictive power and transparency.

In summary, understanding these strengths and limitations not only informs model selection but also guides the implementation of ML-based credit scoring systems that are accurate, fair, and operationally feasible.

7.3 RQ3: What metrics are used to evaluate machine learning credit scoring models?

The evaluation of ML models in credit scoring relies on a variety of metrics, as summarized in Sect. 5 and Table 3. Accuracy remains the most commonly reported metric, reflecting its intuitive appeal. However, its reliability is limited in imbalanced datasets, which are common in credit scoring. Metrics such as AUC, F1-score, G-Mean, and KS are frequently used to address this limitation, providing a more balanced assessment of model performance and better reflecting the ability to correctly identify risky borrowers.

The diversity of metrics reported across studies indicates both the complexity of credit scoring tasks and the absence of standardized evaluation protocols, making direct comparisons across models challenging. In practice, combining multiple metrics is essential to capture different aspects of performance, including overall predictive accuracy, minority-class detection, and discriminatory power. These trends suggest that researchers and practitioners recognize the importance of comprehensive evaluation to ensure models are both accurate and reliable in real-world credit risk assessment.

7.4 RQ4: What are the emerging trends and advances in ML models for credit scoring?

The field of credit scoring has seen significant trends and advancements with the integration of ML techniques that promise to reshape traditional paradigms. These innovations seek to augment the predictive capabilities and efficiency of credit scoring models, ultimately striving towards more inclusive and robust financial practices promoting more inclusive and robust financial practices.

7.4.1 Use of alternative data in credit scoring

Using alternative data in credit scoring is an emerging trend that enhances predictive accuracy by incorporating non-traditional data sources (Niu et al. 2019). Instead of relying solely on traditional financial data, alternative data allows for the analysis of diverse information from platforms such as Facebook, Twitter, and Google, as well as from mobile phone usage data. Psychometric assessments are emerging as new tools for evaluating borrower creditworthiness through alternative credit scoring models. This trend is particularly valuable for borrowers lacking sufficient financial history or considered too risky by traditional models. It provides lenders with alternative indicators to assess borrower behavior and creditworthiness.

These trends reflect a dual emphasis in current credit scoring research: improving predictive accuracy with novel data and models, while ensuring transparency and interpretability for regulatory compliance and ethical lending.

Various studies have demonstrated the effectiveness of using alternative data sources such as social media, mobile phone usage, and psychometric assessments to improve credit scoring models.

De Cnudde et al. (2019) enhanced traditional credit scoring models by incorporating Facebook data. They categorized relationships into Look-A-likes (LAL), friends, and Best Friends Forever (BFF). BFFs displayed stronger predictive value than friends, while interest-driven LAL data surpassed social network data showing the promising future of facebook data in microfinance.

Yu et al. (2020) addressed "real but false data" challenges in credit assessment using douban's social media data. Their criteria-driven data cleansing, including activity and network ratios, led to significant credit score rank changes. This work contributed to trustworthy credit evaluation from social media data, helping to mitigate risks in the internet finance sector.

Niu et al. (2019) investigated using borrowers' mobile phone-derived social network data for loan default prediction. LR and ML methods (RF, AdaBoost, LightGBM) confirmed the significant correlation of the data with loan default and its potential to enhance credit scoring accuracy.

Kulkarni and Dhage (2019) introduced the "Information Trustworthiness" credit scoring system to fuse legacy and emotional/social scores. By leveraging social media interactions and reliable data sources, this system enhances accuracy. The resulting advanced credit score considers personality traits effectively distinguishing default history and non-users, surpassing traditional methods. The use of these alternative datasets aligns with the evolution of credit scoring methodologies reflecting the progressive nature of the advancements in the field.

For the use of mobile phone data, Shema (2019) demonstrated that precise credit scoring models could be built using airtime recharge data, a less invasive option for the privacy of borrowers. Testing against traditional models using loan data, their approach performed equally well and suggested the potential for digital lenders to enhance credit scoring while respecting privacy.

Ots et al. (2020) highlighted the practicality of using mobile phone usage data for credit scoring with a small dataset of 2,503 customers. Employing diverse classification methods, they achieved a 62% AUC in predicting payment behavior. This approach is particularly valuable for smaller companies lacking access to extensive datasets.

Óskarsdóttir et al. (2020) proposed a smartphone-based microlending as a means to enhance financial inclusion. Their approach refined credit scoring models by engineering user data into pseudo-social networks and merging network analysis and ML. Ethical considerations guided this methodology and offered the potential to elevate micro-lending app performance and extend global financial access.

Typically, psychometrics is used to evaluate mental, behavioral, and cognitive traits. The rationale behind incorporating them into credit scoring lies in their potential to reveal certain personality characteristics that might offer insights into an individual's propensity to fulfill financial obligations (Djeundje et al. 2021).

Sifrain (2020) examined the Entrepreneurial Financial Lab (EFL) psychometric test credit scoring model in microfinance using Sogesol. They found the EFL tool outperformed the existing model of Sogesol, though the psychometric model showed limited predictive power. Introducing a new credit scoring model with socio-economic and behavioral factors improved performance. They suggested enhancing credit risk management potential at Sogesol.

Rabecca et al. (2018) investigated adding psychometric testing to the credit scoring system, alongside demographic factors. Their Indonesian case study with PT. Amartha mikro fintek revealed improved credit risk prediction using a combined approach. The psychological test was implemented efficiently, with completion possible within five minutes. Background factors from borrowers and company officers influenced testing time.

Djeundje et al. (2021) assessed predictive accuracy with alternative data for credit risk assessment. Combining email, psychometric, and demographic variables outperformed using demographics alone. This enhanced accuracy supports using email and demographic data when credit history is absent. Similar outcomes were observed with psychometric data. Despite variable results from different sample splits, the approach remains promising. Their study also explored email usage as a predictor using diverse classifiers and identified alternatives for credit risk prediction when traditional data was scarce.

This innovative approach, driven by advancements in ML and big data analytics, offers promising avenues for enhancing credit risk assessment and fostering financial inclusion. The use of alternative data sources such as social media, mobile phone data, and psychometric assessments reflects the evolution of credit scoring methodologies in response to advances in financial technology.

These findings imply that alternative data is most valuable when traditional credit history is sparse or absent, offering lenders a pathway to expand financial inclusion. However, the benefits are not uniform across all contexts: improvements depend heavily on data quality, platform penetration, and the stability of behavioral signals over time. Moreover, reliance on social media or psychometric variables raises concerns about privacy, fairness, and potential proxy discrimination, making careful governance and regulatory oversight essential. In practice, this means that while alternative data can enhance predictive accuracy, it should be integrated as a complement rather than a replacement for traditional financial indicators, with clear safeguards for consent, transparency, and ethical use.

7.4.2 Explainability and interpretability of machine learning models for credit scoring

Interpretability and explainability are critical emerging trends and advancements in the development of ML models for credit scoring. Although often used interchangeably, these terms have distinct meanings. Interpretability refers to the ability to understand model predictions without fully analyzing the internal mechanisms. Explainability, on the other hand, involves clarifying the internal workings of these models in human-understandable terms.

These concepts have gained attention due to the necessity for transparency in ML models. The need arises from the desire to understand how and why a particular borrower is granted or denied a loan (Bussmann et al. 2021). Additionally, the proprietary nature and increasing complexity of these models make it difficult for specialists to understand their inner workings.

Consequently, developing tools that can explain these models in a reliable and interpretable manner has become crucial. LIME and SHAP are two methods used to explain predictions of black-box models. LIME uses locally accurate explanations for individual predictions based on the assumption of local linearity. SHAP values explain predictions by quantifying the marginal contribution of each feature to the prediction.

Bussmann et al. (2021) proposed an interpretable AI model for credit risk management, especially in peer-to-peer lending. Using correlation networks and Shapley values, the model groups AI predictions by shared explanations. Analyzing 15,000 small businesses, the study found that similar financial characteristics could explain and predict credit scores for both risky and non-risky borrowers.

Bücker et al. (2022) proposed a framework for enhancing the interpretability of credit scoring models to align with transparency, auditability, and explainability goals for "black box" ML models. By using methods such as LIME and SHAP, they showed that interpretability comparable to traditional scorecards can be achieved while preserving predictive capabilities.

Ayari and Guetari (2025) emphasized the importance of interpretability in credit scoring models, showing that while EL improves predictive accuracy, understanding the role of individual features and base learners is crucial for transparency and trust. Using SHAP, their approach quantified the contribution of each feature and classifier, supporting regulatory compliance and ethical decision-making.

Bussmann et al. (2021) utilized LIME and SHAP to explain ML-based credit scoring models on the lending club dataset. Local and global insights were gained using these techniques and SHAP kernel comparisons were explored. The results proved that LIME and SHAP provided consistent explanations that are in line with financial logic.

By embracing these trends, the credit scoring domain is making an impressive move toward merging the power of ML with comprehensibility. Thus, the goal is to balance data availability with building trust among regulators by ensuring ML models are transparent and understandable.

The broader implication of these advances is that explainability has shifted from a technical option to a regulatory and operational requirement. By making complex models intelligible to credit officers, auditors, and borrowers, tools such as SHAP and LIME not only foster trust but also help detect instability or bias in model predictions. This means that the successful deployment of ML in credit scoring now depends as much on interpretability and accountability as on predictive performance. Future developments will likely need to balance model complexity with explainability, ensuring that accuracy gains are not achieved at the expense of transparency or compliance.

7.5 RQ5: What are the challenges in adopting ML models for credit scoring?

The adoption of ML models in credit scoring presents numerous challenges, despite their potential to significantly enhance predictive accuracy and efficiency. Among these challenges, three critical issues stand out: interpretability, potential biases, and the curse of dimensionality. These aspects are critical to ensuring models are not only effective but also equitable and transparent. In this subsection, we delve into these challenges.

7.5.1 Interpretability

Interpretability continues to pose a significant challenge in applying complex ML models to credit scoring. Although these models can offer high predictive accuracy, their decision-making processes are often opaque, which makes it difficult for lenders to understand how credit decisions are made. This lack of transparency can be problematic for borrowers and regulators who require clear explanations for credit outcomes. Thus, achieving a balance between high interpretability and predictive performance is therefore a complex task.

Techniques such as LIME and SHAP have been developed to enhance the interpretability of black-box models. While these methods offer valuable insights into model predictions, they can sometimes provide inconsistent explanations for similar input data and may not fully uncover underlying model biases. Additionally, these techniques can be susceptible to adversarial attacks (Slack et al. 2020), raising concerns about their reliability in sensitive applications like credit scoring. Failure to address this challenge may result in difficulties aligning model outputs with regulatory requirements and industry standards, potentially leading to a lack of trust in credit scoring systems.

Consequently, the need to develop and implement effective techniques for model interpretability remains a pressing concern in the field of credit scoring. These techniques should not only make the decision-making process of ML models more understandable but also ensure that critical insights are clear in a way that aids in informed credit risk assessment.

7.5.2 Potential biases

Biases in training datasets can lead ML models to exhibit unfairness based on several criteria. In credit scoring, for example, these criteria may include age, gender, race, caste, and religion, among others. Therefore, addressing bias is a major challenge in developing ML models, as they can perpetuate existing disparities if not carefully designed. In credit scoring, failure to address bias can lead to serious consequences, including the reinforcement of social inequalities and ethical concerns (Ahmed 2022).

To mitigate biases, various techniques can be employed, including pre-processing, in-processing, and post-processing methods (Jammalamadaka and Itapu 2023). Pre-processing techniques involve manipulating the training data to reduce bias before model training. In-processing techniques can be integrated directly into the learning process to adjust model parameters during training to eliminate bias. Post-processing techniques refine model outputs to meet fairness criteria after initial predictions are made. Although mitigating bias remains complex, these proactive measures can significantly enhance the fairness and reliability of credit scoring systems.

7.5.3 The curse of dimensionality

The curse of dimensionality presents significant challenges when applying ML to high-dimensional data. As the number of dimensions increases, computational demands rise, resulting in longer training times and greater memory consumption. Sparse data in high-dimensional spaces can lead to overfitting as models struggle to generalize. Models are more likely to learn noise and irrelevant patterns, which adversely affect accuracy. Similarly, distance-based metrics become less meaningful, affecting algorithms such as clustering and

KNN. High-dimensional data increases model variability, complicating performance evaluation and necessitating careful hyperparameter tuning (Jia et al. 2022).

Effective feature selection methods can help address these challenges. Techniques such as filter, wrapper, and embedded approaches are used to reduce the number of features while maintaining model generalization (Laborda and Ryoo 2021). Filter methods assess individual features based on statistical measures like correlation, information gain, or chi-square tests to identify those with greater predictive power. Wrapper methods evaluate different feature subsets using a chosen ML model, such as forward selection, backward elimination, or recursive feature elimination. While these methods provide a more exhaustive search by considering the interaction between features, they can be computationally expensive for large feature spaces. Embedded methods integrate feature selection into the model training process, optimizing feature relevance as part of the learning algorithm. By applying appropriate feature selection techniques, the impact of the curse of dimensionality can be managed to enhance the robustness of ML models.

7.5.4 Behavioral and attitudinal data integration

In addition to technical and data-related challenges, recent research has highlighted the importance of borrowers' attitudes and behavioral factors in credit risk assessment. According to a qualitative study on educational loan repayment by postgraduate students in India (Bhandary et al. 2023), loan repayment is influenced by both ability and willingness to repay. While ability is commonly measured through financial indicators, willingness is reflected in attitudinal dimensions such as credit history, debt utility, financial knowledge, prioritizing repayment, and integrity. The study also identified gratification, debt burden, and lifestyle preferences as negative attitudes associated with delinquency. Integrating such subjective dimensions into ML models remains a significant challenge due to the difficulty in collecting and quantifying such data. However, doing so could improve the predictive power and fairness of credit scoring systems, especially in contexts like student or microfinance loans where attitudinal variables play a more pronounced role.

7.5.5 Synthesis of challenges and practical implications

Overall, these challenges illustrate that while ML models can enhance predictive performance, practical adoption in credit scoring requires careful attention to interpretability, fairness, and feature management. Addressing interpretability is crucial for regulatory compliance and maintaining borrower trust. Mitigating bias ensures equitable lending decisions and aligns with ethical standards. Managing high-dimensional data through feature selection not only improves computational efficiency but also reduces overfitting and enhances generalization. Integrating behavioral and attitudinal data, although complex, can significantly improve model accuracy and inclusiveness, especially for non-traditional borrower segments such as students or microfinance clients. Together, these considerations highlight that successful ML deployment in credit scoring is as much about responsible model design and data handling as it is about predictive accuracy.

7.6 Comparison with existing literature reviews

This SLR differs from and extends prior surveys in several important ways. Existing reviews, such as Dastile et al. (2020), focused primarily on the performance of individual ML and ensemble classifiers from 2010 to 2018. Their review identified ensemble classifiers like RFs and XGBoost as outperforming single models, and highlighted CNNs as the leading DL architecture. Our results confirm these observations, particularly the consistent high performance of ensemble and hybrid models across benchmark datasets. However, our review includes a broader range of ensemble and hybrid techniques, such as GA-NN and multi-stage ensemble models, which demonstrated even higher accuracy and AUC in recent studies.

This study (Kumar et al. 2021) presented a SLR focused on credit scoring within rural finance, emphasizing how fintech and AI technologies are transforming credit assessment in underserved areas. Their review highlighted the limitations of traditional banking in rural contexts and the benefits of integrating ML models such as ANN, SVM, RF, and hybrid approaches to improve financial inclusion. While their findings emphasized the socio-economic impact and regulatory considerations, their performance analysis remained largely conceptual. In contrast, our review adopts a broader technical scope, systematically comparing model performance across standard benchmark datasets and incorporating recent advancements such as model interpretability (SHAP, LIME) and alternative data sources (social media, mobile phone usage, psychometrics). Therefore, our work complements by providing detailed empirical evidence and expanding the applicability of ML credit scoring models beyond rural-focused contexts

Hayashi (2022) emphasized the dominance of Deep Belief Networks and CNNs, and discussed the challenges of interpretability and the potential of transforming structured data into image-like formats. In contrast, our findings reveal that while DL methods do show promise, they remain less commonly applied than ensemble or hybrid methods, largely due to their complexity and lower interpretability in practice.

Lenka et al. (2022) focused on ensemble models for imbalanced data and highlighted the use of SMOTE and GA for feature selection. Our results corroborate this trend by showing that models using SMOTE, GAs, and ensemble classifiers tend to achieve higher performance, especially on imbalanced datasets such as Lending Club and German credit data. Additionally, we found that the integration of GA with NN or boosting further improves performance, which was not explored in their review.

Other surveys such as Markov et al. (2022) and Kamimura et al. (2023) provided historical and methodological overviews, highlighting the shift from traditional models (like LR, DTs) to hybrid and DL approaches. Our review confirms this evolution, but also emphasizes new developments such as the use of alternative data (social media, mobile usage, psychometrics) and the growing importance of explainability techniques like SHAP and LIME, which were not widely discussed in previous SLRs.

Moreover, previous reviews often lacked detailed, ranked comparisons of model performance across datasets. Our review contributes uniquely in this regard by compiling ranked tables of accuracy and AUC on four widely used datasets. This offers a clearer picture of the comparative effectiveness of different models and configurations.

In summary, while our findings align with prior work in recognizing the value of ensemble and hybrid models for credit scoring, we extend existing literature by incorporating

more recent developments, evaluating a wider set of techniques and datasets, and offering practical comparisons across metrics and model types. We also address emerging themes like interpretability, alternative data, and algorithmic fairness, which are increasingly central to responsible credit scoring.

A recent study by Bhandary and Ghosh (2025) offered a comprehensive empirical comparison of both traditional and modern ML techniques for credit scoring, using a well known real world dataset. Their work compared models such as LR, linear discriminant analysis, SVMs, RF, XGBoost, and deep neural networks, and showed that DL models achieved the highest predictive performance in terms of F1-score, G-mean, and AUC. Their results confirmed that, while statistical methods like LR still offered competitive accuracy, modern ML approaches (particularly deep neural networks) consistently outperformed traditional techniques across several performance metrics.

Notably, the study also addressed interpretability concerns in complex models by analyzing feature importance within the deep neural network. It found that behavioral factors such as age and payment history were key predictors, reinforcing earlier literature emphasizing the role of personal characteristics and repayment behavior in default prediction. Although the study did not specifically address credit utilization, recent literature (Bhandary and Ghosh 2025) has emphasized its critical role in dynamic scoring, suggesting that high utilization by reliable borrowers may actually signal profitability for lenders. Furthermore, while Bhandary et al. discussed LGD in the context of aligning scoring with risk based capital planning, complementary work has stressed the importance of incorporating IFRS 9 standards holistically, particularly PD, EAD, and LGD, to ensure compliance and financial soundness in advanced scoring models.

8 Limitations of this review

Firstly, this study included peer-reviewed journal papers and conference articles related to credit scoring. Our initial search filtration strategy yielded a large number of articles. Several non-relevant studies were subsequently identified and excluded. This process ensured that the selected research papers met the inclusion criteria of the study. Nonetheless, including more related papers would have enriched our conclusions.

Secondly, the search for articles was confined to only four online databases: IEEE Xplore, ACM Digital Library, Springer Link, and Google Scholar. Nonetheless, there may be other digital libraries with relevant studies that were overlooked. Moreover, identifying all relevant research published within the five-year scope of our investigation proved challenging due to the increasing volume of studies in the field of credit scoring using ML techniques. Despite this limitation, our exhaustive exploration provides valuable insights into the current landscape of credit scoring research employing ML methodologies.

Finally, this review lacks a comparative analysis to identify the most effective models for credit scoring. This challenge stems from the fact that the articles reviewed often used different evaluation metrics, even when relying on common datasets. As a result, any comparison would be subjective and potentially unreliable.

9 Conclusion

This SLR aimed to provide a comprehensive analysis of ML applications in financial credit scoring from 2018 to 2024. By examining 63 carefully selected studies, the review identified the major ML methodologies employed, evaluated their strengths and limitations, and highlighted current trends and challenges in the domain.

The key findings indicate that ensemble and hybrid models, which often combine feature optimization and multiple classifiers, consistently outperform traditional single models in terms of accuracy and discrimination power across popular credit scoring datasets. DL techniques show promise with large datasets but face limitations related to interpretability and data availability. Moreover, the integration of alternative data sources such as social media, mobile usage, and psychometric data is an emerging trend that can enhance credit scoring, particularly for borrowers lacking formal credit histories. The growing emphasis on explainability methods such as LIME and SHAP demonstrates the field's commitment to transparency and regulatory compliance.

The implications of these findings are significant for researchers, practitioners, and financial institutions. ML offers improved predictive capabilities and operational efficiencies, but challenges around model interpretability, bias mitigation, and computational complexity remain barriers to widespread adoption. Addressing these issues is critical for building trustworthy, fair, and scalable credit scoring systems that promote financial inclusion.

This review is limited by the heterogeneity of datasets, evaluation metrics, and methodologies in the literature, which complicates direct performance comparisons. Additionally, while this SLR highlights key trends and challenges, the fast-evolving nature of ML means continuous updates are necessary to capture new advances.

Future research should focus on establishing standardized benchmarking protocols to enable fair and consistent evaluation of credit scoring models. Investigating robust integration of alternative data with privacy safeguards, developing more interpretable and bias-aware models, and exploring scalable solutions suitable for diverse institutional contexts are promising directions. Emphasis on real-world deployment challenges and ethical considerations will also be essential to translate research advances into practical credit risk management tools.

Acknowledgements The authors extend their appreciation to the Deanship of Scientific Research at King Khalid University for funding this work through a large group Research Project under grant number RGP2/428/46.

Author contributions H.A. (Helmi Ayari) was responsible for writing the manuscript, conducting the research, and analyzing the data. H.A. also prepared the initial draft and final version of the paper. R.G. (Pr. Ramzi Guetari) provided critical guidance throughout the research process, including overseeing the evaluation of results and suggesting necessary revisions. N.K. (Pr. Naoufel Kraiem) reviewed the final manuscript for additional insights and feedback, and assisted with financial aspects related to the project, leveraging his affiliation with the funding laboratory. All authors contributed to the revision of the manuscript and approved the final version.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Acharya S, Pustokhina IV, Pustokhin DA, Geetha BT, Joshi GP, Nebhen J, Yang E, Seo C (2022) An improved gradient boosting tree algorithm for financial risk management. Knowl Manag Res Pract 20(4):543–554. https://doi.org/10.1080/14778238.2021.1954489

Adisa J, Ojo S, Owolawi P, Pretorius A, Ojo SO (2022) Credit score prediction using genetic algorithm-LSTM technique. In: 2022 Conference on information communications technology and society (ICTAS), pp 1–6. https://doi.org/10.1109/ICTAS53252.2022.9744714 . IEEE

Ahmed F (2022) Ethical aspects of artificial intelligence in banking. J Res Econ Finance Manag 1(2):55–63. https://doi.org/10.56596/jrefm.v1i2.7

Aji NA, Dhini A (2019) Credit scoring through data mining approach: a case study of mortgage loan in Indonesia. In: 2019 16th International conference on service systems and service management (ICSSSM), pp 1–5. https://doi.org/10.1109/ICSSSM.2019.8887731

Ala'raj M, Abbod MF, Majdalawieh M (2021) Modelling customers credit card behaviour using bidirectional LSTM neural networks. J Big Data 8(1):1–27. https://doi.org/10.1186/s40537-021-00461-7

Ala'raj M, Abbod MF, Majdalawieh M, Jum'a L (2022) A deep learning model for behavioural credit scoring in banks. Neural Comput Appl 34(8):5839–5866. https://doi.org/10.1007/s00521-021-06695-z

Ampountolas A, Nde TN, Date P, Constantinescu C (2021) A machine learning approach for micro-credit scoring. Risks 9(3):50. https://doi.org/10.3390/risks9030050

Aniceto MC, Barboza F, Kimura H (2020) Machine learning predictivity applied to consumer creditworthiness. Future Bus J 6(1):1–14. https://doi.org/10.1186/s43093-020-00041-w

Ariza-Garzón MJ, Arroyo J, Caparrini A, Segovia-Vargas M-J (2020) Explainability of a machine learning granting scoring model in peer-to-peer lending. IEEE Access 8:64873–64890. https://doi.org/10.1109/ACCESS.2020.2984412

Atiya AF (2001) Bankruptcy prediction for credit risk using neural networks: a survey and new results. IEEE Trans Neural Netw 12(4):929–935. https://doi.org/10.1109/72.935101

Ayari H, Guetari R (2025) Integrating genetic algorithms and ensemble learning for improved and transparent credit scoring. In: International conference on business information systems, pp 225–238. https://doi.org/10.1007/978-3-031-94193-1_17 . Springer

Bai M, Zheng Y, Shen Y (2022) Gradient boosting survival tree with applications in credit scoring. J Oper Res Soc 73(1):39–55. https://doi.org/10.1080/01605682.2021.1919035

Bao W, Ning L, Yue K (2019) Integration of unsupervised and supervised machine learning algorithms for credit risk assessment. Expert Syst Appl 128:301–315. https://doi.org/10.1016/j.eswa.2019.02.033

Bhandary R, Ghosh BK (2025) Credit card default prediction: an empirical analysis on predictive performance using statistical and machine learning methods. J Risk Financ Manag 18(1):23. https://doi.org/10.3390/jrfm18010023

Bhandary R, Shenoy SS, Shetty A, Shetty AD (2024) Education loan repayment: a systematic literature review. J Financ Serv Market 29(4):1365–1376. https://doi.org/10.1057/s41264-023-00248-2

Bhandary R, Shenoy SS, Shetty A, Shetty AD (2023) Attitudes toward educational loan repayment among college students: a qualitative enquiry. J Financ Counseling Plan 34(2). https://doi.org/10.1891/JFCP-2022-0069

Bhatore S, Mohan L, Reddy YR (2020) Machine learning techniques for credit risk evaluation: a systematic literature review. J Bank Financ Technol 4(1):111–138. https://doi.org/10.1007/s42786-020-00020-3

Boughaci D, Alkhawaldeh A, Jaber JJ, Hamadneh N (2021) Classification with segmentation for credit scoring and bankruptcy prediction. Empir Econ 61:1281–1309. https://doi.org/10.1007/s00181-020-01901-8

Bradford M (2007) Personal credit information: privacy and information security issues—the experian view. Bus Inf Rev 24(4):253–256. https://doi.org/10.1177/0266382107084893

Breiman L (2001) Random forests. Mach Learn 45(1):5–32. https://doi.org/10.1023/A:1010933404324

Bücker M, Szepannek G, Gosiewska A, Biecek P (2022) Transparency, auditability, and explainability of machine learning models in credit scoring. J Oper Res Soc 73(1):70–90. https://doi.org/10.1080/01605682.2021.1922098

Bussmann N, Giudici P, Marinelli D, Papenbrock J (2021) Explainable machine learning in credit risk management. Comput Econ 57:203–216. https://doi.org/10.1007/s10614-020-10042-0

Cao NT, Tran LH, Ton-That AH (2021) Using machine learning to create a credit scoring model in banking and finance. In: 2021 IEEE Asia-Pacific conference on computer science and data engineering (CSDE), pp 1–5. https://doi.org/10.1109/CSDE53843.2021.9718414

Caruana R, Niculescu-Mizil A (2006) An empirical comparison of supervised learning algorithms. In: Proceedings of the 23rd international conference on machine learning. ICML '06, pp. 161–168. Association for Computing Machinery, New York. https://doi.org/10.1145/1143844.1143865

Cervantes J, Garcia-Lamont F, Rodríguez-Mazahua L, Lopez A (2020) A comprehensive survey on support vector machine classification: applications. Challenges Trends Neurocomput 408:189–215. https://doi.org/10.1016/j.neucom.2019.10.118

Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP (2002) Smote: synthetic minority over-sampling technique. J Artif Intell Res 16(1):321–357. https://doi.org/10.1613/jair.953

Chen X, Li S, Xu X, Meng F, Cao W (2020) A novel GSCI-based ensemble approach for credit scoring. IEEE Access 8:222449–222465. https://doi.org/10.1109/ACCESS.2020.3043937

Dastile X, Celik T (2021) Making deep learning-based predictions for credit scoring explainable. IEEE Access 9:50426–50440. https://doi.org/10.1109/ACCESS.2021.3068854

Dastile X, Celik T, Potsane M (2020) Statistical and machine learning models in credit scoring: a systematic literature survey. Appl Soft Comput 91:106263. https://doi.org/10.1016/j.asoc.2020.106263

De Cnudde S, Moeyersoms J, Stankova M, Tobback E, Javaly V, Martens D (2019) What does your facebook profile reveal about your creditworthiness? Using alternative data for microfinance. J Oper Res Soc 70(3):353–363. https://doi.org/10.1080/01605682.2018.1434402

Diaconescu P, Neagoe V-E (2020) Credit scoring using deep learning driven by optimization algorithms. In: Proceedings of the 2020 12th international conference on electronics, computers and artificial intelligence (ECAI), pp. 1–6. https://doi.org/10.1109/ECAI50035.2020.9223139

Dike HU, Zhou Y, Deveerasetty KK, Wu Q (2018) Unsupervised learning based on artificial neural network: a review. In: 2018 IEEE International conference on cyborg and bionic systems (CBS), pp 322–327. https://doi.org/10.1109/CBS.2018.8612259

Djeundje VB, Crook J, Calabrese R, Hamid M (2021) Enhancing credit scoring with alternative data. Expert Syst Appl 163:113766. https://doi.org/10.1016/j.eswa.2020.113766

Dm O, Mm M (2018) Comparison of accuracy of support vector machine model and logistic regression model in predicting individual loan defaults. Am J Appl Math Stat 6(6):266–271 (https://pubs.sciepub.com/ajams/6/6/8/index.html)

Dong X, Yu Z, Cao W, Shi Y, Ma Q (2020) A survey on ensemble learning. Front Comput Sci 14(2):241–258. https://doi.org/10.1007/s11704-019-8208-z

Dumitrescu E, Hué S, Hurlin C, Tokpavi S (2022) Machine learning for credit scoring: improving logistic regression with non-linear decision-tree effects. Eur J Oper Res 297(3):1178–1192. https://doi.org/10.1016/j.ejor.2021.06.053

ElKelish WW (2021) The international financial reporting standards 9 financial instruments, information quality and stock returns in the modern technology era. J Appl Acc Res 22(3):465–483. https://doi.org/10.1108/JAAR-12-2019-0164

Frank D, Bhandary R, Prabhu SK (2024) Higher education loan schemes across the globe: a systematic review on the utility derived and burden associated with educational debt. J Risk Financ Manag 17(12):566. https://doi.org/10.3390/jrfm17120566

Friedman JH (2001) Greedy function approximation: a gradient boosting machine. Ann Stat 29(5):1189–1232. https://doi.org/10.1214/aos/1013203451

Friedman N, Geiger D, Goldszmidt M (1997) Bayesian network classifiers. Mach Learn 29(2):131–163. https://doi.org/10.1023/A:1007465528199

Gicic A, Subasi A (2019) Credit scoring for a microcredit data set using the synthetic minority oversampling technique and ensemble classifiers. Expert Syst 36(2):12363. https://doi.org/10.1111/exsy.12363

Goh RY, Lee LS, Seow HV, Gopal K (2020) Hybrid harmony search-artificial intelligence models in credit scoring. Entropy 22(9):989. https://doi.org/10.3390/e22090989

Gu J, Wang Z, Kuen J, Ma L, Shahroudy A, Shuai B, Liu T, Wang X, Wang G, Cai J (2018) Recent advances in convolutional neural networks. Pattern Recogn 77:354–377. https://doi.org/10.1016/j.patcog.2017.10.013

Guetari R, Ayari H, Sakly H (2023) Computer-aided diagnosis systems: a comparative study of classical machine learning versus deep learning-based approaches. Knowl Inf Syst 65(10):3881–3921. https://doi.org/10.1007/s10115-023-01894-7

Guo S, He H, Huang X (2019) A multi-stage self-adaptive classifier ensemble model with application in credit scoring. IEEE Access 7:78549–78559. https://doi.org/10.1109/ACCESS.2019.2922676

Guo G, Wang H, Bell D, Bi Y, Greer K (2003) KNN model-based approach in classification. In: On the move to meaningful internet systems 2003: CoopIS, DOA, and ODBASE: OTM confederated international conferences, CoopIS, DOA, and ODBASE 2003, Catania, Sicily, Italy, November 3–7, 2003. Proceedings, pp 986–996. https://doi.org/10.1007/978-3-540-39964-3_62

Hand DJ (2009) Measuring classifier performance: a coherent alternative to the area under the roc curve. Mach Learn 77(1):103–123. https://doi.org/10.1007/s10994-009-5119-5

Hayashi Y (2022) Emerging trends in deep learning for credit scoring: a review. Electronics 11(19):3181. https://doi.org/10.3390/electronics11193181

He H, Zhang W, Zhang S (2018) A novel ensemble method for credit scoring: adaption of different imbalance ratios. Expert Syst Appl 98:105–117. https://doi.org/10.1016/j.eswa.2018.01.012

Hochreiter S (1998) The vanishing gradient problem during learning recurrent neural nets and problem solutions. Int J Uncertain Fuzziness Knowl-Based Syst 6(2):107–116. https://doi.org/10.1142/S0218488598000094

Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Hosmer DW, Lemeshow S (2000) Applied logistic regression, 3rd edn. Wiley, New York. https://doi.org/10.1002/0471722146

Huang CL, Chen MC, Wang CJ (2007) Credit scoring with a data mining approach based on support vector machines. Expert Syst Appl 33(4):847–856. https://doi.org/10.1016/j.eswa.2006.07.007

Ibrahim A, Olagunju SO (2022) Improving credit scoring performance using two-stage technique. Abacus (Mathematics Science Series) 49(2):329 (https://www.man-nigeria.org.ng/issues/ABA-SCI-2022-35.pdf)

Jammalamadaka KR, Itapu S (2023) Responsible ai in automated credit scoring systems. AI Ethics 3(2):485–495. https://doi.org/10.1007/s43681-022-00175-3

Jia W, Sun M, Lian J, Hou S (2022) Feature dimensionality reduction: a review. Complex Intell Syst 8(3):2663–2693. https://doi.org/10.1007/s40747-021-00637-x

Jiao W, Hao X, Qin C (2021) The image classification method with cnn-xgboost model based on adaptive particle swarm optimization. Information 12(4):156. https://doi.org/10.3390/info12040156

Jin Y, Liu Y, Zhang W, Zhang S, Lou Y (2021) A novel multi-stage ensemble model with multiple k-means-based selective undersampling: an application in credit scoring. J Intell Fuzzy Syst 40(5):9471–9484. https://doi.org/10.3233/JIFS-201954

Jin Y, Zhang W, Wu X, Liu Y, Hu Z (2021) A novel multi-stage ensemble model with a hybrid genetic algorithm for credit scoring on imbalanced data. IEEE Access 9:143593–143607. https://doi.org/10.1109/ACCESS.2021.3120086

Kamimura ES, Pinto ARF, Nagano MS (2023) A recent review on optimization methods applied to credit scoring models. J Econ Finance Admin Sci 28(56):352–371. https://doi.org/10.1108/JEFAS-09-2021-0193

Kazemi HR, Khalili-Damghani K, Sadi-Nezhad S (2021) Tuning structural parameters of neural networks using genetic algorithm: a credit scoring application. Expert Syst 38(7):12733. https://doi.org/10.1111/exsy.12733

Kazemi HR, Khalili-Damghani K, Sadi-Nezhad S (2023) Estimation of optimum thresholds for binary classification using genetic algorithm: an application to solve a credit scoring problem. Expert Syst 40(3):13203. https://doi.org/10.1111/exsy.13203

Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, Ye Q, Liu T (2017) LightGBM: a highly efficient gradient boosting decision tree. In: Proceedings of the 31st international conference on neural information processing systems. NIPS'17, vol. 30, pp 3149–3157. Curran Associates, Inc., Red Hook, NY. https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Kenny C (2018) The equifax data breach and the resulting legal recourse. Brooklyn J Corporate Financ Commercial Law 13(1):215–238 (https://heinonline.org/HOL/LandingPage?handle=hein.journals/broojcfc13&div=14 &id= &page=)

Khedr MH, Azim NA, Ammar AM (2021) A new prediction approach for preventing default customers from applying personal loans using machine learning. Int J Comput Sci Mob Comput 10(12):71–82. https://doi.org/10.47760/ijcsmc.2021.v10i12.009

Kulkarni SV, Dhage SN (2019) Advanced credit score calculation using social media and machine learning. J Intell Fuzzy Syst 36(3):2373–2380. https://doi.org/10.3233/JIFS-169948

Kumar A, Sharma S, Mahdavi M (2021) Machine learning (ml) technologies for digital credit scoring in rural finance: a literature review. Risks 9(11):192. https://doi.org/10.3390/risks9110192

Laborda J, Ryoo S (2021) Feature selection in a credit scoring model. Mathematics 9(7):746. https://doi.org/10.3390/math9070746

Lenka SR, Bisoy SK, Priyadarshini R, Sain M (2022) Empirical analysis of ensemble learning for imbalanced credit scoring datasets: a systematic review. Wirel Commun Mob Comput 2022(1):6584352. https://doi.org/10.1155/2022/6584352

Li G, Ma H, Liu R, Shen M, Zhang K (2021) A two-stage hybrid default discriminant model based on deep forest. Entropy 23(5):582. https://doi.org/10.3390/e23050582

Li H, Qiu H, Sun S, Chang J, Tu W (2022) Credit scoring by one-class classification driven dynamical ensemble learning. J Oper Res Soc 73(1):181–190. https://doi.org/10.1080/01605682.2021.1944824

Liu W, Fan H, Xia M (2021) Multi-grained and multi-layered gradient boosting decision tree for credit scoring. Appl Intell 51(15):10643–10661. https://doi.org/10.1007/s10489-021-02715-6

Loo WT, Khaw KW, Chew X, Alnoor A, Lim ST (2023) Predicting the loan default using machine learning algorithms: a case study in India. J Eng Technol 14(2):17–27 (https://jet.utem.edu.my/jet/article/view/6346)

Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. In: Advances in neural information processing systems 30 (NeurIPS 2017), pp 4765–4774. https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Macey JR, Miller GP (1988) Trans union reconsidered. Yale Law J 98(1):127–143

Maharjan M (2022) Comparative analysis of data mining methods to analyze personal loans using decision tree and Naïve Bayes classifier. Int J Educ Manag Eng 12(4):33. https://doi.org/10.5815/ijeme.2022.04.04

Malmi T (2001) Balanced scorecards in finnish companies: a research note. Manag Account Res 12(2):207–220. https://doi.org/10.1006/mare.2000.0154

Markov A, Seleznyova Z, Lapshin V (2022) Credit scoring methods: latest trends and points to consider. J Finance Data Sci 8:180–201. https://doi.org/10.1016/j.jfds.2022.07.002

Moscato V (2021) A benchmark of machine learning approaches for credit score prediction. Expert Syst Appl 165:113986. https://doi.org/10.1016/j.eswa.2020.113986

Mukid MA, Widiharih T, Rusgiyono A, Prahutama A (2018) Credit scoring analysis using weighted k nearest neighbor. J Phys: Conf Ser, vol. 1025, p 012114. https://doi.org/10.1088/1742-6596/1025/1/012114. IOP Publishing

Nalic J, Martinovic G, Žagar D (2020) New hybrid data mining model for credit scoring based on feature selection algorithm and ensemble classifiers. Adv Eng Inform 45:101130. https://doi.org/10.1016/j.aei.2020.101130

Neagoe V, Ciotec AD, Cucu GS (2018) Deep convolutional neural networks versus multilayer perceptron for financial prediction. In: 2018 International conference on communications (COMM), pp 201–206. https://doi.org/10.1109/ICComm.2018.8484751. IEEE

Niu B, Ren J, Li X (2019) Credit scoring using machine learning by combining social network information: evidence from peer-to-peer lending. Information 10(12):397. https://doi.org/10.3390/info10120397

Noble WS (2006) What is a support vector machine? Nat Biotechnol 24(12):1565–1567. https://doi.org/10.1038/nbt1206-1565

Nobre J, Neves RF (2019) Combining principal component analysis, discrete wavelet transform and XGBoost to trade in the financial markets. Expert Syst Appl 125:181–194. https://doi.org/10.1016/j.eswa.2019.01.083

Óskarsdóttir M, Bravo C, Sarraute C, Baesens B, Vanthienen J (2020) Credit scoring for good: enhancing financial inclusion with smartphone-based microlending. arXiv:2001.10994

Ots H, Liiv I, Tur D (2020) Mobile phone usage data for credit scoring. Databases and information systems: 14th International Baltic conference, DBIS 2020, Tallinn, Estonia, June 16–19, 2020, Proceedings 14, 82–95 https://doi.org/10.1007/978-3-030-57672-1_7

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Moher D (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372:71. https://doi.org/10.1136/bmj.n71

Parvin AS, Saleena B (2020) An ensemble classifier model to predict credit scoring–comparative analysis. In: 2020 IEEE international symposium on smart electronic systems (iSES) (Formerly iNiS), pp 27–30.https://doi.org/10.1109/iSES50453.2020.00017

Pławiak P, Abdar M, Acharya UR (2019) Application of new deep genetic cascade ensemble of SVM classifiers to predict the Australian credit scoring. Appl Soft Comput 84:105740. https://doi.org/10.1016/j.asoc.2019.105740

Pławiak P, Abdar M, Pławiak J, Makarenkov V, Acharya UR (2020) DGHNL: a new deep genetic hierarchical network of learners for prediction of credit scoring. Inf Sci 516:401–418. https://doi.org/10.1016/j.ins.2019.12.045

Pratiwi H, Mukid MA, Hoyyi A, Widiharih T (2019) Credit scoring analysis using pseudo nearest neighbor. J Phys: Conf Ser 1217:012100. https://doi.org/10.1088/1742-6596/1217/1/012100. (**IOP Publishing**)

Rabecca H, Atmaja ND, Safitri S (2018) Psychometric credit scoring in indonesia microfinance industry: a case study in PT Amartha Mikro Fintek. In: Proceedings of the 3rd international conference on management in emerging markets (ICMEM 2018), pp 620–631. ICMEM. https://www.researchgate.net/publication/333809157_Psychometric_Credit_Scoring_in_Indonesia_Microfinance_Industry_A_Case_Study_in_PT_Amartha_Mikro_Fintek

Ribeiro MT, Singh S, Guestrin C (2019) Why Should I trust you?: explaining the predictions of any classifier. ArXiv160204938 Cs Stat https://doi.org/10.1145/2939672.2939778

Rofik R, Aulia R, Musaadah K, Ardyani SSF, Hakim AA (2024) The optimization of credit scoring model using stacking ensemble learning and oversampling techniques. J Inf Syst Explor Res 2(1). https://doi.org/10.52465/joiser.v2i1.203

Rumelhart DE, Hinton GE, Williams RJ (1986) Learning representations by back-propagating errors. Nature 323(6088):533–536. https://doi.org/10.1038/323533a0

Safavian SR, Landgrebe D (1991) A survey of decision tree classifier methodology. IEEE Trans Syst Man Cybern 21(3):660–674. https://doi.org/10.1109/21.97458

Shema A (2019) Effective credit scoring using limited mobile phone data. In: Proceedings of the tenth international conference on information and communication technologies and development, pp 1–11. https://doi.org/10.1145/3287098.3287116

Shen F, Zhao X, Li Z, Li K, Meng Z (2019) A novel ensemble classification model based on neural networks and a classifier optimisation technique for imbalanced credit risk evaluation. Phys A 526:121073. https://doi.org/10.1016/j.physa.2019.121073

Shen F, Zhao X, Kou G, Alsaadi FE (2021) A new deep learning ensemble credit risk evaluation model with an improved synthetic minority oversampling technique. Appl Soft Comput 98:106852. https://doi.org/10.1016/j.asoc.2020.106852

Sifrain R (2020) Does psychometric testing in microfinance actually work? The case of sogesol. J Financ Risk Manag 9(03):278. https://doi.org/10.4236/jfrm.2020.93016

Slack D, Hilgard S, Jia E, Singh S, Lakkaraju H (2020) Fooling LIME and SHAP: adversarial attacks on post hoc explanation methods. In: Proceedings of the AAAI/ACM conference on AI, ethics, and society, pp 180–186. https://doi.org/10.1145/3375627.3375830

Smith BC (2011) Stability in consumer credit scores: level and direction of fico score drift as a precursor to mortgage default and prepayment. J Hous Econ 20(4):285–298. https://doi.org/10.1016/j.jhe.2011.09.001

Suleiman S, Ibrahim A, Usman D, Yabo BI, Muhammad HU (2021) Improving credit scoring classification performance using self-organizing map-based machine learning techniques. Eur J Adv Eng Technol 8(10):28–35 (https://zenodo.org/record/10651297)

Syed Nor SH, Ismail S, Yap BW (2019) Personal bankruptcy prediction using decision tree model. J Econ Finance Admin Sci 24(47):157–170. https://doi.org/10.1108/JEFAS-08-2018-0076

Teles G, Rodrigues JJ, Rabêlo RA, Kozlov SA (2021) Comparative study of support vector machines and random forests machine learning algorithms on credit operation. Softw Pract Exp 51(12):2492–2500. https://doi.org/10.1002/spe.2842

Tokarski M (2020) Protection of individuals in the light of EU regulation 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data. Safety Defense 2:63–74. https://doi.org/10.37105/sd.86

Tran KQ, Duong BV, Tran LQ, Tran AL, Nguyen AT, Nguyen KV (2021) Machine learning-based empirical investigation for credit scoring in Vietnam's banking. In: Advances and trends in artificial intelligence. From theory to practice: 34th international conference on industrial, engineering and other applications of applied intelligent systems (IEA/AIE 2021), Kuala Lumpur, Malaysia, July 26–29, 2021, Proceedings, Part II, pp 564–574. https://doi.org/10.1007/978-3-030-79463-7_48

Tripathi D, Edla DR, Kuppili V, Bablani A, Dharavath R (2018) Credit scoring model based on weighted voting and cluster-based feature selection. Procedia Comput Sci 132:22–31. https://doi.org/10.1016/j.procs.2018.05.055

Tripathi D, Edla DR, Cheruku R, Kuppili V (2019) A novel hybrid credit scoring model based on ensemble feature selection and multilayer ensemble classification. Comput Intell 35(2):371–394. https://doi.org/10.1111/coin.12200

Trivedi SK (2020) A study on credit scoring modeling with different feature selection and machine learning approaches. Technol Soc 63:101413. https://doi.org/10.1016/j.techsoc.2020.101413

Van Eck N, Waltman L (2010) Software survey: Vosviewer, a computer program for bibliometric mapping. Scientometrics 84(2):523–538. https://doi.org/10.1007/s11192-009-0146-3

Wang C, Han D, Liu Q, Luo S (2018) A deep learning approach for credit scoring of peer-to-peer lending using attention mechanism LSTM. IEEE Access 7:2161–2168. https://doi.org/10.1109/ACCESS.2018.2887138

Wang T, Li J (2019) An improved support vector machine and its application in P2P lending personal credit scoring. In: IOP conference series: materials science and engineering, vol 490, p 062041. https://doi.org/10.1088/1757-899X/490/6/062041

Wright RE (1995) Logistic regression. Reading and Understanding Multivariate Statistics, 217–244

Xia Y, He L, Li Y, Fu Y, Xu Y (2021) A dynamic credit scoring model based on survival gradient boosting decision tree approach. Technol Econ Dev Econ 27(1):96–119. https://doi.org/10.3846/tede.2020.13997

Xing Q, Yu C, Huang S, Zheng Q, Mu X, Sun M (2024) Enhanced credit score prediction using ensemble deep learning model. arXiv:2410.00256

Yao J-R, Chen J-R (2019) A new hybrid support vector machine ensemble classification model for credit scoring. J Inf Technol Res 12(1):77–88. https://doi.org/10.4018/JITR.2019010106

Yotsawat W, Wattuya P, Srivihok A (2021) A novel method for credit scoring based on cost-sensitive neural network ensemble. IEEE Access 9:78521–78537. https://doi.org/10.1109/ACCESS.2021.3083490

Yotsawat W, Wattuya P, Srivihok A (2021) Improved credit scoring model using XGBoost with Bayesian hyper-parameter optimization. Int J Electr Comput Eng 11(6):5477. https://doi.org/10.11591/ijece.v11i6.pp5477-5487

Yu X, Yang Q, Wang R, Fang R, Deng M (2020) Data cleaning for personal credit scoring by utilizing social media data: an empirical study. IEEE Intell Syst 35(2):7–15. https://doi.org/10.1109/MIS.2020.2972214

Yuan K, Chi G, Zhou Y, Yin H (2022) A novel two-stage hybrid default prediction model with k-means clustering and support vector domain description. Res Int Bus Financ 59:101536. https://doi.org/10.1016/j.ribaf.2021.101536

Zhang T, Chi G (2021) A heterogeneous ensemble credit scoring model based on adaptive classifier selection: an application on imbalanced data. Int J Financ Econ 26(3):4372–4385. https://doi.org/10.1002/ijfe.2019

Zhang S, Li X, Zong M, Zhu X, Cheng D (2017) Learning k for knn classification. ACM Trans Intell Syst Technol 8(3):1–19. https://doi.org/10.1145/2990508

Zhang Z, Niu K, Liu Y (2020) A deep learning based online credit scoring model for P2P lending. IEEE Access 8:177307–177317. https://doi.org/10.1109/ACCESS.2020.3027337

Zhang W, Yang D, Zhang S, Ablanedo-Rosas JH, Wu X, Lou Y (2021) A novel multi-stage ensemble model with enhanced outlier adaptation for credit scoring. Expert Syst Appl 165:113872. https://doi.org/10.1016/j.eswa.2020.113872

Zhang W, Yang D, Zhang S (2021) A new hybrid ensemble model with voting-based outlier detection and balanced sampling for credit scoring. Expert Syst Appl 174:114744. https://doi.org/10.1016/j.eswa.2021.114744

Zhang X, Yang Y, Zhou Z (2018) A novel credit scoring model based on optimized random forest. In: 2018 IEEE 8th annual computing and communication workshop and conference (CCWC), pp 60–65. https://doi.org/10.1109/CCWC.2018.8301707

Zhu B, Yang W, Wang H, Yuan Y (2018) A hybrid deep learning model for consumer credit scoring. In: Proceedings of the 2018 international conference on artificial intelligence and big data (ICAIBD), pp 205–208. https://doi.org/10.1109/ICAIBD.2018.8396195

Zou Y, Gao C (2022) Extreme learning machine enhanced gradient boosting for credit scoring. Algorithms 15(5):149. https://doi.org/10.3390/a15050149

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.